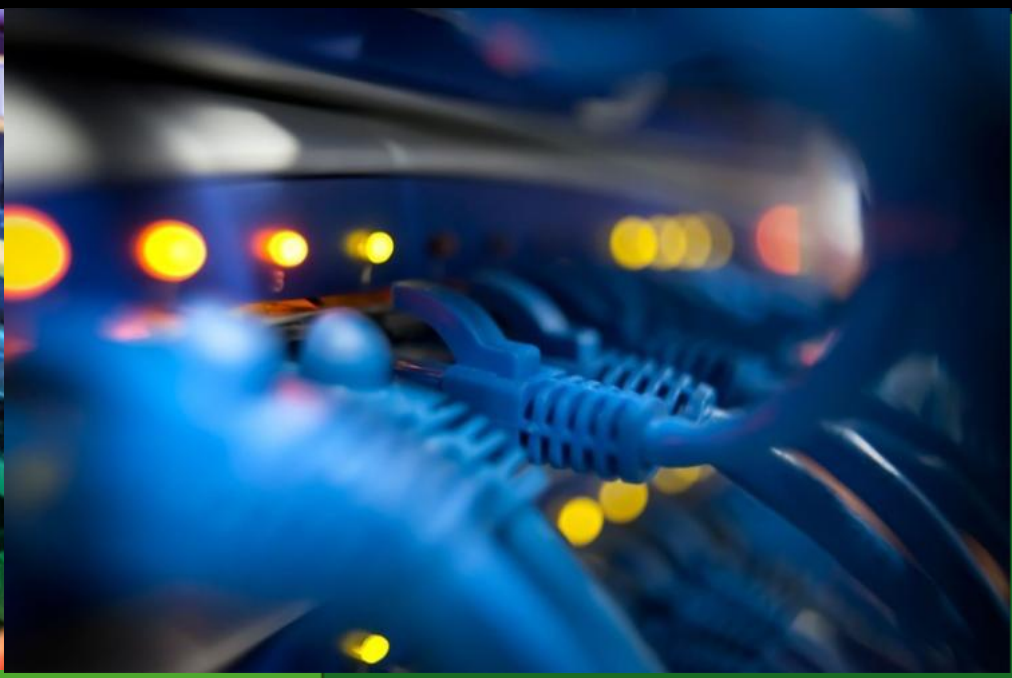
AWS Lab: Configuring IAM, VPC, and EC2



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Purpose

These labs were designed and finished to explore the capabilities of AWS and further our knowledge into how AWS IAM, VPC, and AWS EC2 work and function together inside of an AWS Architecture.

Background

Cloud computing, as defined by the **International Organization for Standardization (ISO)** as "a paradigm for enabling network access to a scalable and elastic pool of shareable physical or virtual resources with self-service provisioning and administration on demand". It is commonly called **"the cloud."**

The cloud is a recent invention and has many differences and usages in comparison to on-premises networking. For example, **the cloud is on-demand**, meaning you could use computing capabilities at any time, with any amount, without requiring any human interaction with a service provider. Practically, this shifts the traditional on-premises networking that has immense capital/up-front expenditure **to a pay-as-you-go model with operational expenditures.**

The cloud also maintains a global premise, which allows you to deploy anywhere in the world within a few clicks. The immense number of data centers across the globe leased by the Cloud Platform (In this case, AWS), also ensures redundancy and high availability, whereas on premises would require duplicating hardware, which duplicates cost.

The cloud can be analogous to housing. Whereas buying one house can be expensive, and not fit your needs, renting a house of scalable size, which meets your needs, with a lease controlled majorly by you would grant scalability and availability. Despite this, on-premises still has benefits over cloud computing such as having ownership of hardware which leads to cost predictability at scale, no shared-tenancy concerns, and in general, a full autonomy over hardware and infrastructure.

Amazon Web Services (AWS) is a subsidiary of Amazon which provides said on-demand cloud computing on a metered, pay-as-you-go basis. One of the foundational services of AWS is Amazon EC2, which allows users to deploy a virtual cluster of computers with high availability. These virtual computers emulate most of the functionalities of an actual computer and are rented.

Amazon Identity and Access Management (IAM) is a web service that helps you control access to your AWS resources. Using IAM allows you to manage the permissions that control which AWS resources users can access. IAM allows you to authenticate and

authorize people to use resources and provides said infrastructure.

IAM has different ways to enforce this, including IAM Policies, IAM Roles, IAM Groups, and IAM User. An IAM User represents an individual who uses AWS and has long-term credentials. By default, they have zero permissions. IAM Users may be collectively added into an IAM Group, who share the same permissions. Instead of attaching an IAM Policy to each user, you can attach it to a group. An AWS IAM Policy is a JSON document that explicitly defines what actions are allowed and not allowed on AWS. Like a user, an IAM role is similar, but is designed to be assumed temporarily for whoever needs it. It's commonly used to grant AWS services permission to interact with other AWS services.

Amazon Elastic Block Storage (EBS) is a service that allows you to create "Storage Volumes" and attach them to Amazon EC2 instances. Once attached, you can create a file system on top of these volumes, run a database, or use them just like you would use block storage. Amazon EBS volumes are placed in a specific Availability Zone, where they are automatically replicated to protect from the failure of a single component. All EBS volume types offer durable snapshot capabilities and are designed for high availability.

While traditionally, object storage is best for unstructured data and analytics, and file storage is best for shared access/enterprise file workloads, Block Storage is best for data bases, boot volumes, and low-latency workloads. And as the name suggests, while file storage stores data in accessible, hierarchical files/folders, block storage breaks the data into blocks that are stored independently, and it appears as a raw disk attached to a virtual machine.

Databases can be divided into two common categories, relational and non-relational. Relational databases organize information into tables with rows and columns and mostly use SQL to manage and access data. Oftentimes these tables will interact with other tables and are very ideal for transactional systems. Non-relational databases are more flexible, and store data in a variety of methods, including documents, key-value pairs, graphs, and more. Databases provide reliable, durable storage, which is needed because of their responsibilities, such as reading, writing, accessing, and managing data.

Scaling in Cloud Computing refers to changing compute capabilities based on need. Vertical Scaling would be to change the size or power of a single server by increasing its capabilities, such as more CPU, more RAM, more storage. As for horizontal scaling, it is to distribute a workload across an increased number of servers. Autoscaling in Cloud Computing is a feature that automatically adjusts computing resources based on the demand set. Instead of manually adding or removing compute resources, an auto scaler can dynamically scale or descale your infrastructure in real

time based on the limits you set on it. For example, when a CPU threshold is met, new EC2 instances can be launched automatically and then terminated when not needed.

Auto scaling is oftentimes combined with load balancing, which distributes workload evenly across all servers/instances to ensure that one server isn't being overworked and provides proper redundancy. Overall, monitoring, auto scaling, and load balancing work together to increase an architecture's fault tolerance, reliability, availability, and cost optimization.

Lab Summary

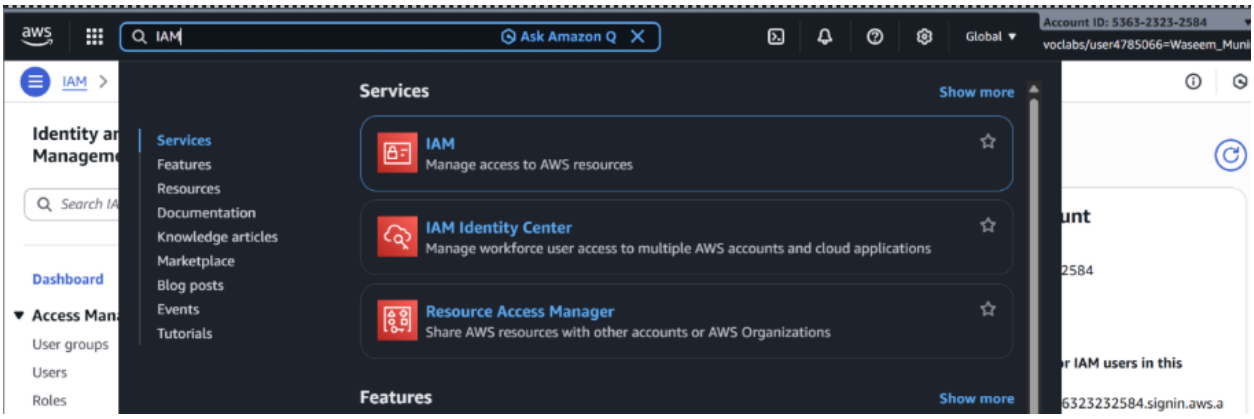
In this lab we configured three different sub-labs. The first involves setting up AWS IAM and exploring the capabilities of security management. The second involves setting up a Virtual Private Cloud (VPC) and launching a web server. In the third, we set up an Amazon EC2 Instance and explore its capabilities.

Configurations

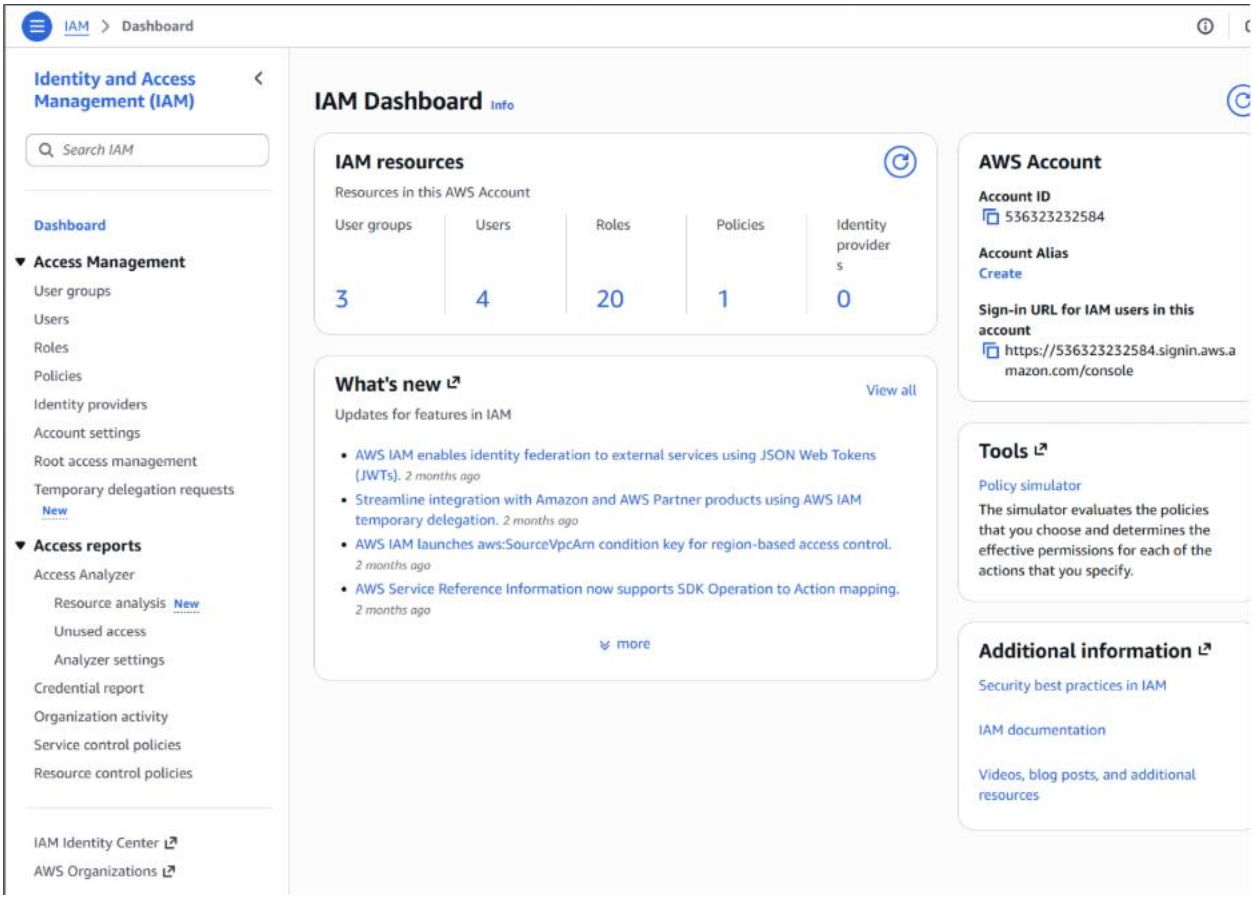
Lab Part 1: Configure AWS IAM

Note: *All labs were completed within the labs set up by the AWS Academy Cloud Foundations course and ideally should be done through there for all labs to properly work.*

1. Click on the navbar, and search for IAM. Click on the first service that appears.

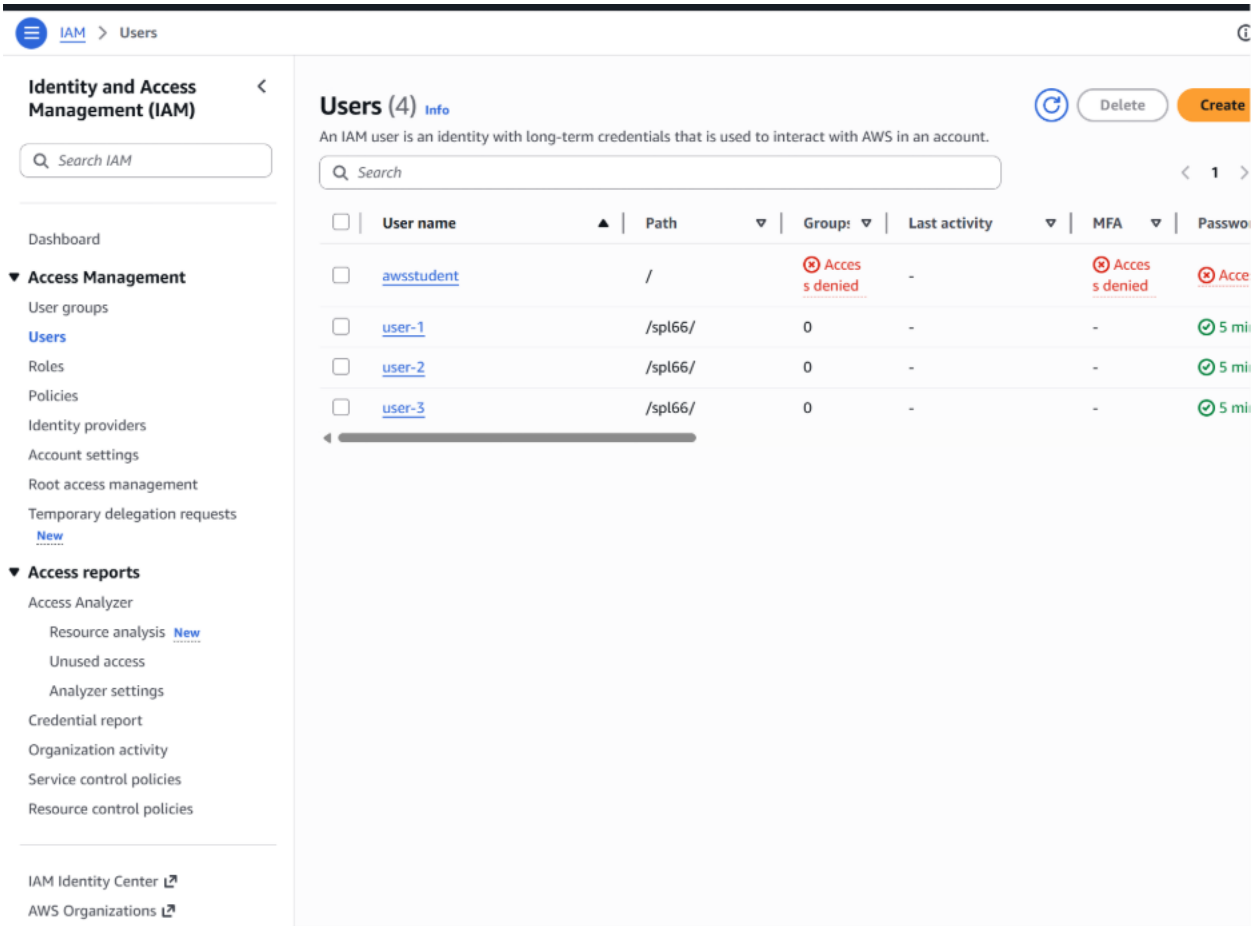


2. Choose "Users" on the left



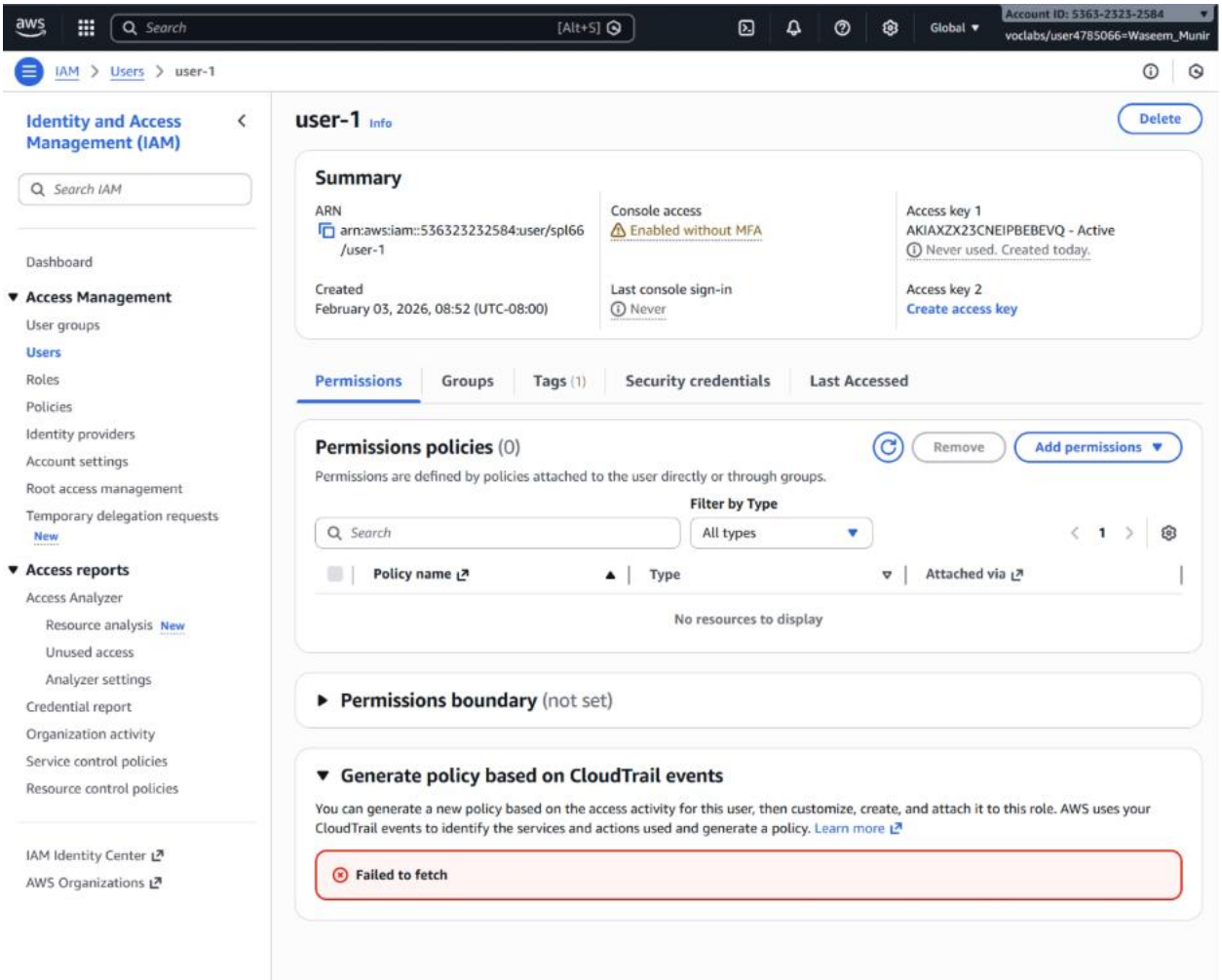
3. Choose the “user-1” link

Note: *Notice that user-1 does not have any permissions.*



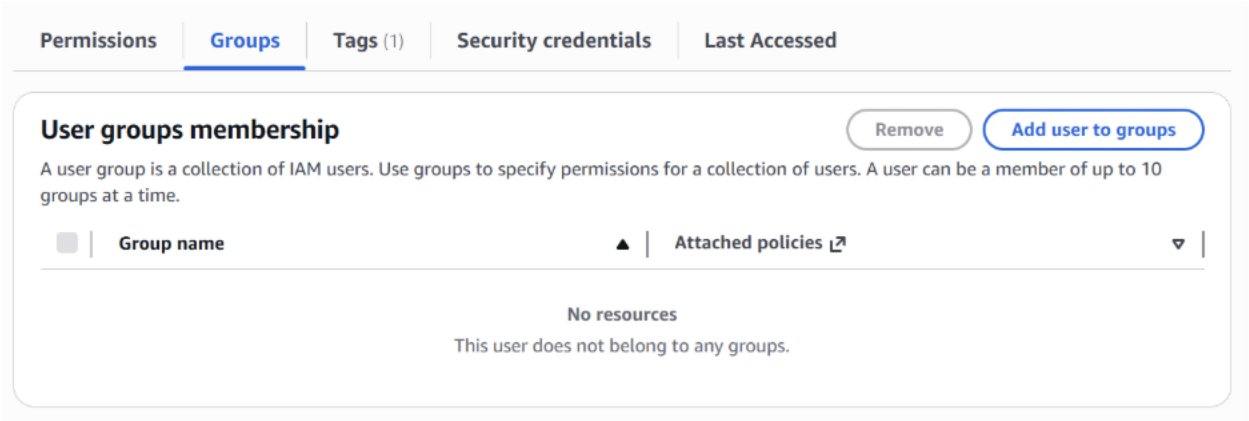
4. Choose the Groups Tab

Note: *user-1 is not a member of any groups.*

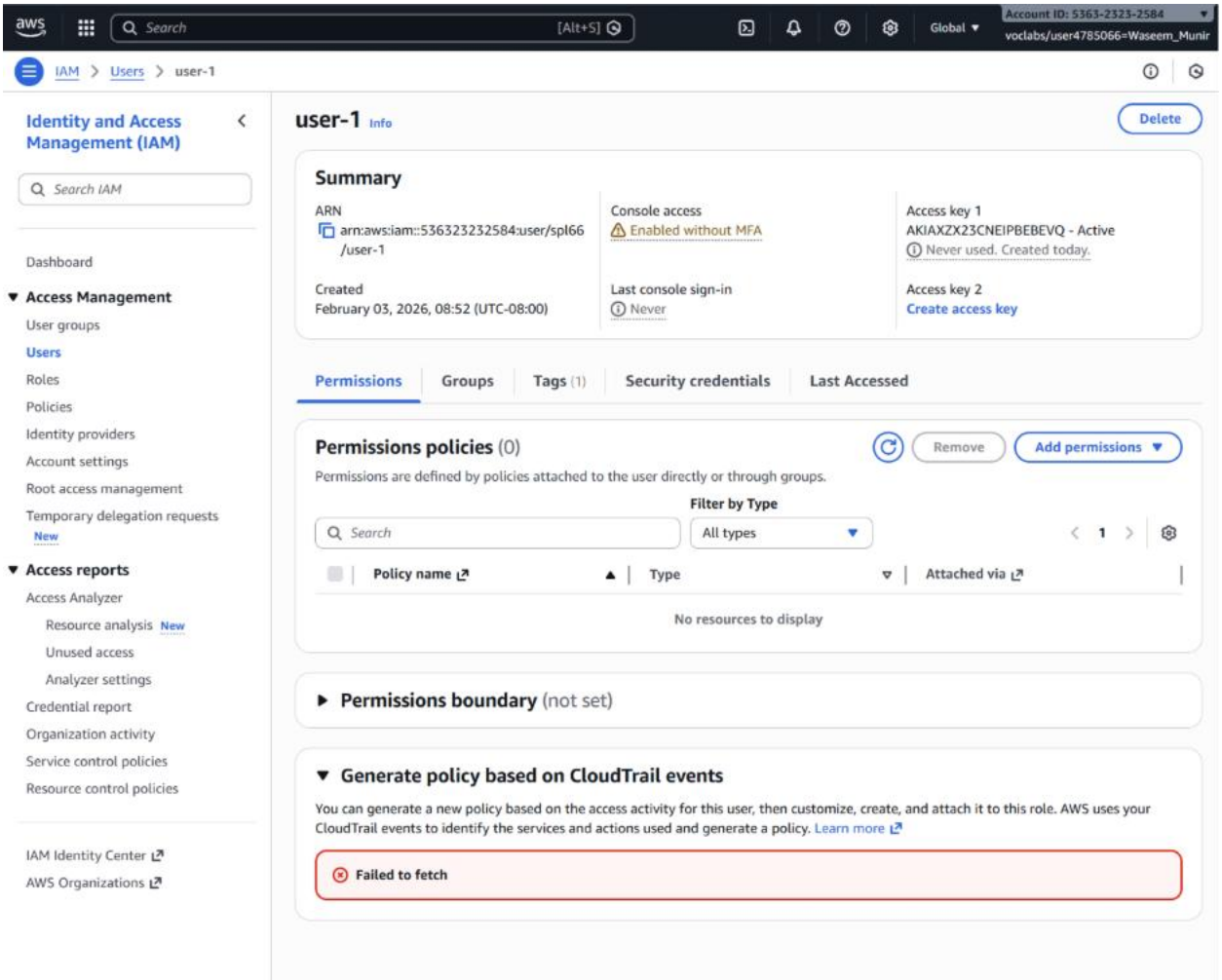


5. Choose the “Security Credentials” tab

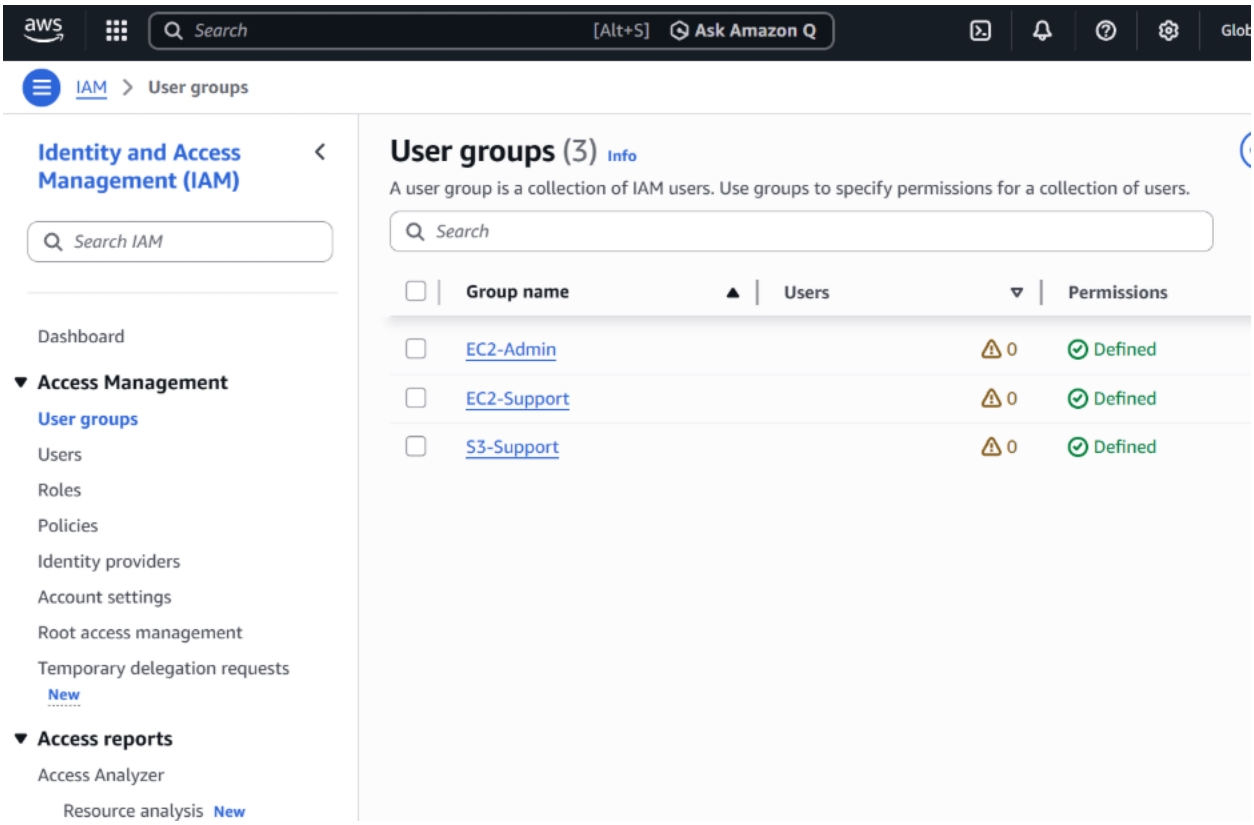
Note: *user-1 is assigned a Console password.*



6. Choose “User Groups”



7. Choose “EC2-Support” group link



8. Choose the "Permissions" tab & click the (+) icon next to the ReadOnly policy

Note: *This policy grants permissions to Get and List resources in Amazon S3.*

Note: *A policy defines actions allowed or denied for specific AWS resources.*

Explanatory Note:

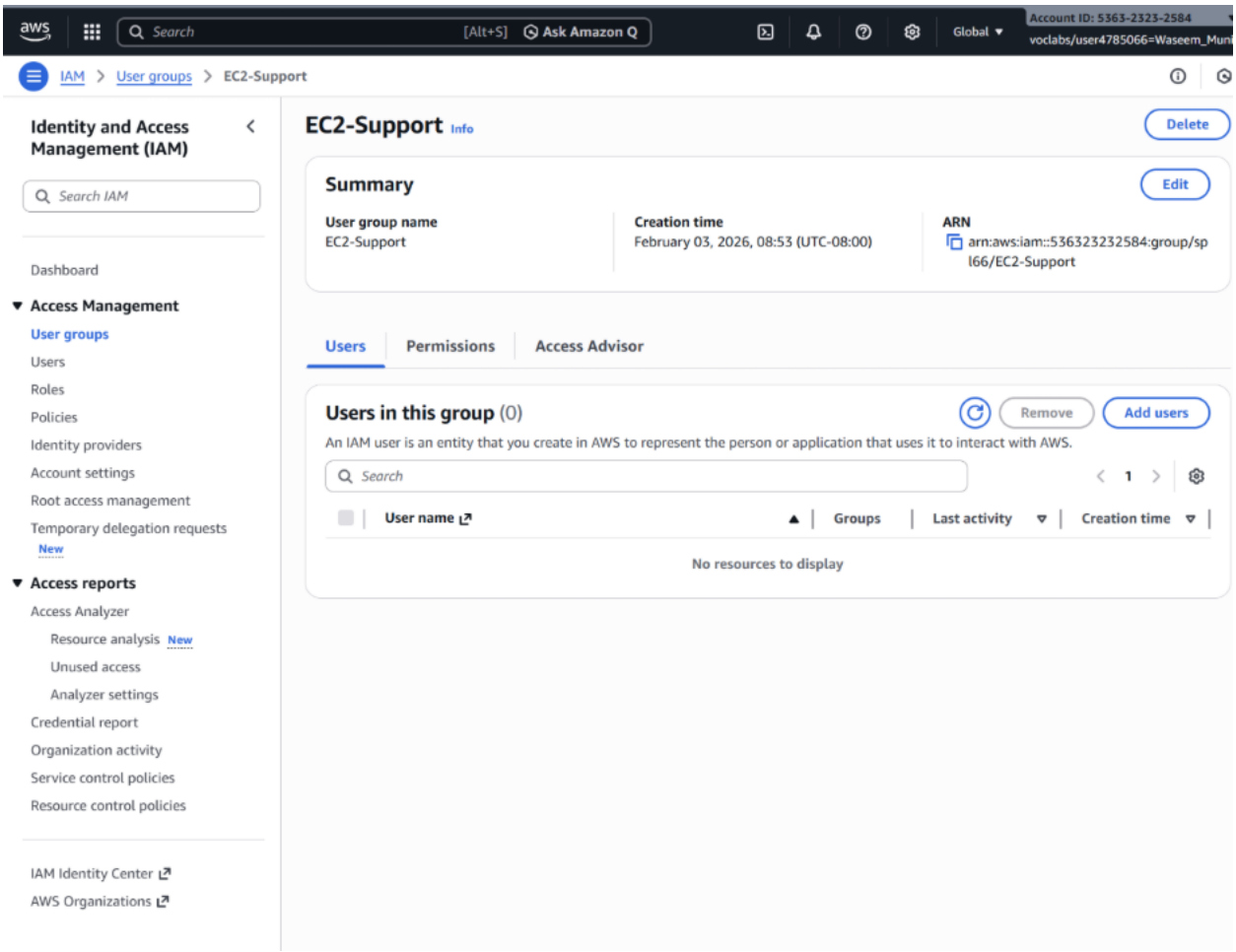
This policy is granting permission to List and Describe information about EC2, Elastic Load Balancing, CloudWatch and Auto Scaling. This ability to view resources, but not modify them, is ideal for assigning a Support role.

The basic structure of the statements in an IAM Policy is:

Effect says whether to Allow or Deny the permissions.

Action specifies the API calls that can be made against an AWS Service (e.g., cloudwatch:ListMetrics).

*Resource defines the scope of entities covered by the policy rule (e.g., a specific Amazon S3 bucket or Amazon EC2 instance, or * which means any resource).*


aws
Search
[Alt+S]
Ask Amazon Q
Global
Account ID: 5363-2323-2584
voclabs/user4785066=Waseem_Muni
IAM > User groups > EC2-Support
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests
New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Resource control policies
IAM Identity Center
AWS Organizations
EC2-Support Info
Delete
Summary
Edit
User group name
EC2-Support
Creation time
February 03, 2026, 08:53 (UTC-08:00)
ARN
arn:aws:iam::536323232584:group/spl66/EC2-Support
Users
Permissions
Access Advisor
Users in this group (0)
Remove
Add users
An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.
Search
1
User name
Groups
Last activity
Creation time
No resources to display

EC2-Support Info

Delete

Summary

Edit

User group name
EC2-Support

Creation time
February 03, 2026, 08:53 (UTC-08:00)

ARN
arn:aws:iam::536323232584:group/spl66/EC2-Support

Users | Permissions | Access Advisor

Permissions policies (1) Info

Simulate | Remove | Add permissions

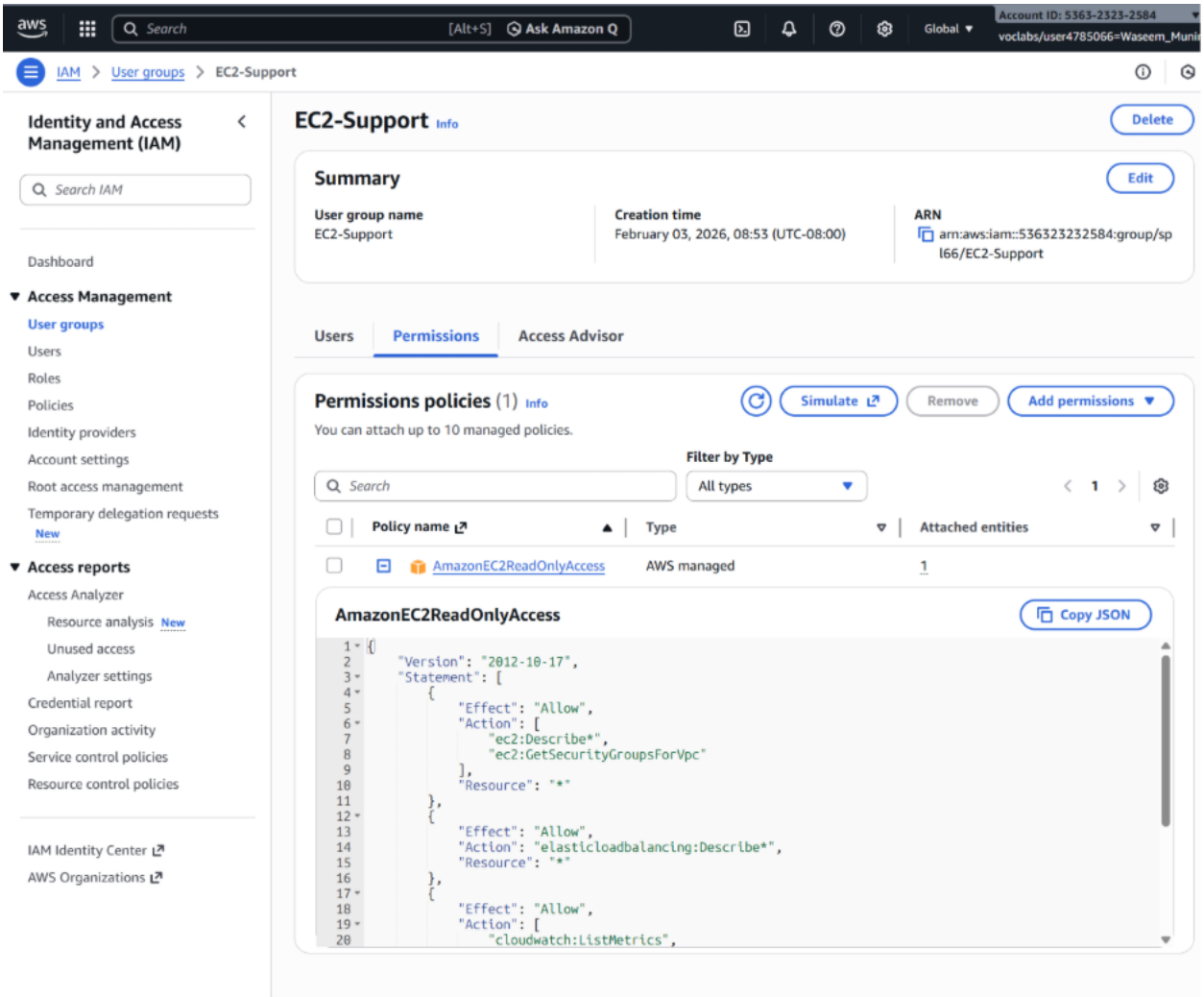
You can attach up to 10 managed policies.

Filter by Type

Search | All types

1

Policy name	Type	Attached entities
AmazonEC2ReadOnlyAccess	AWS managed	1



9. Go back to “User Groups”

10. Choose the “S3-Support” group link and choose “Permissions”

Note: *This policy grants permissions to Get and List resources in Amazon S3.*

11. Click the (+) icon next to the ReadOnly policy to read it, and (-) to close it.

12. Go back to “User Groups”

13. Choose the EC2-Admin group link & click on “Permissions”

Note: *This Group is slightly different from the other two. Instead of a Managed Policy, it has an Inline Policy, which is a policy assigned to just one User or Group. Inline Policies are used to apply permissions for one-off situations.*

14. Click the (+) icon to open the policy details and (-) to close it.

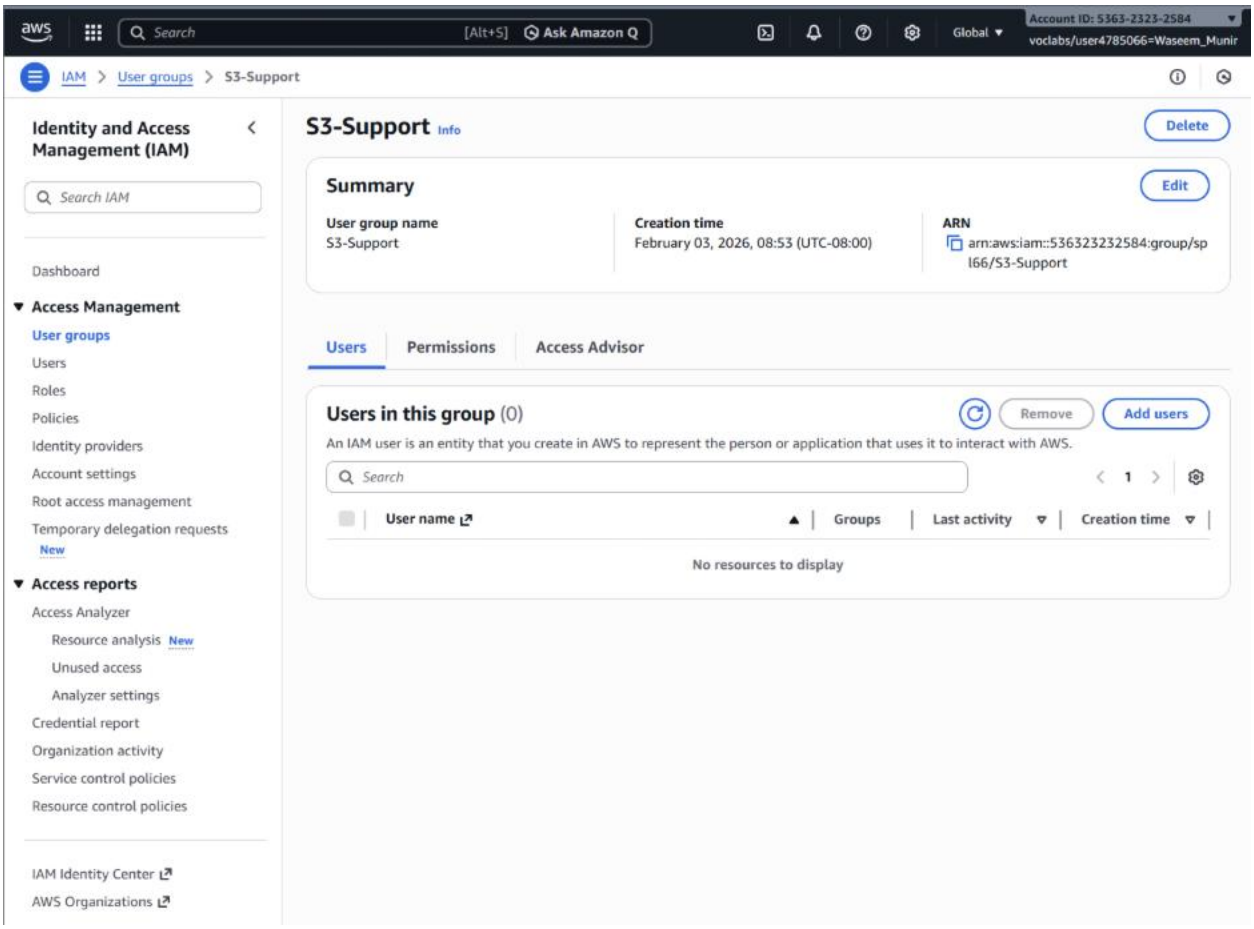
Note: *This policy grants permission to view (Describe) information about Amazon EC2 and the ability to Start and Stop instances.*

Hypothetical Lab Business Scenario

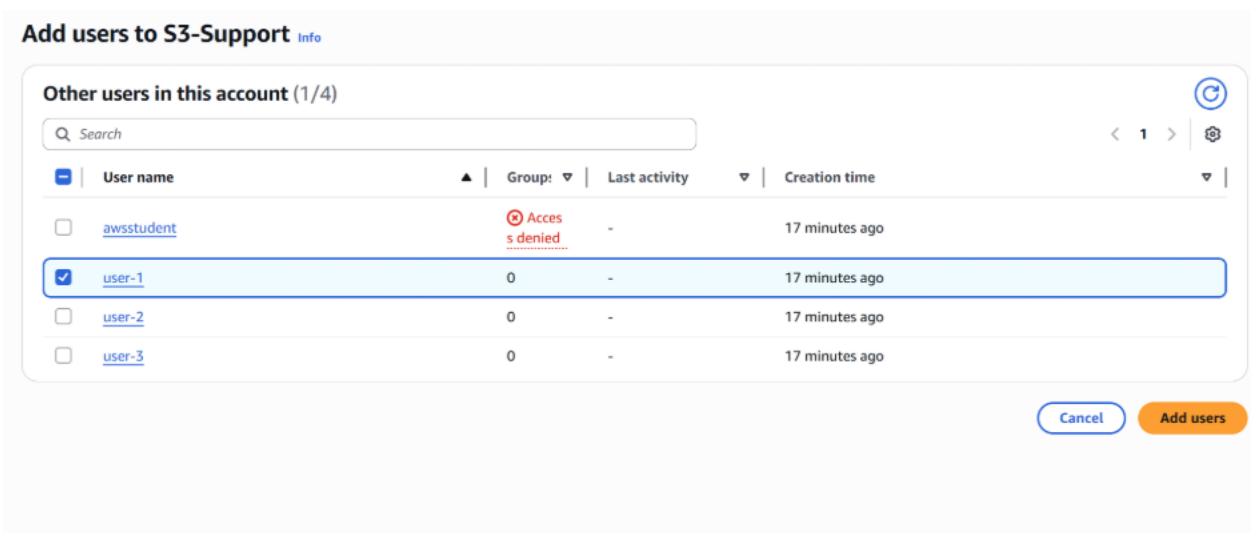
Your company is growing its use of Amazon Web Services and is using many Amazon EC2 instances and a great deal of Amazon S3 storage. You wish to give access to new staff depending upon their job function:

User	In-Group	Permissions
user-1	S3-Support	Read-Only access to S3
user-2	EC2-Support	Read-Only access to EC2
user-3	EC2-Admin	View, Start, Stop EC2 Instances

15. Choose “User Groups,” then “S3-Support”, then “Users”, then “Add Users”



16. Configure the following
    - Select “user-1”
    - Choose “Add Users”

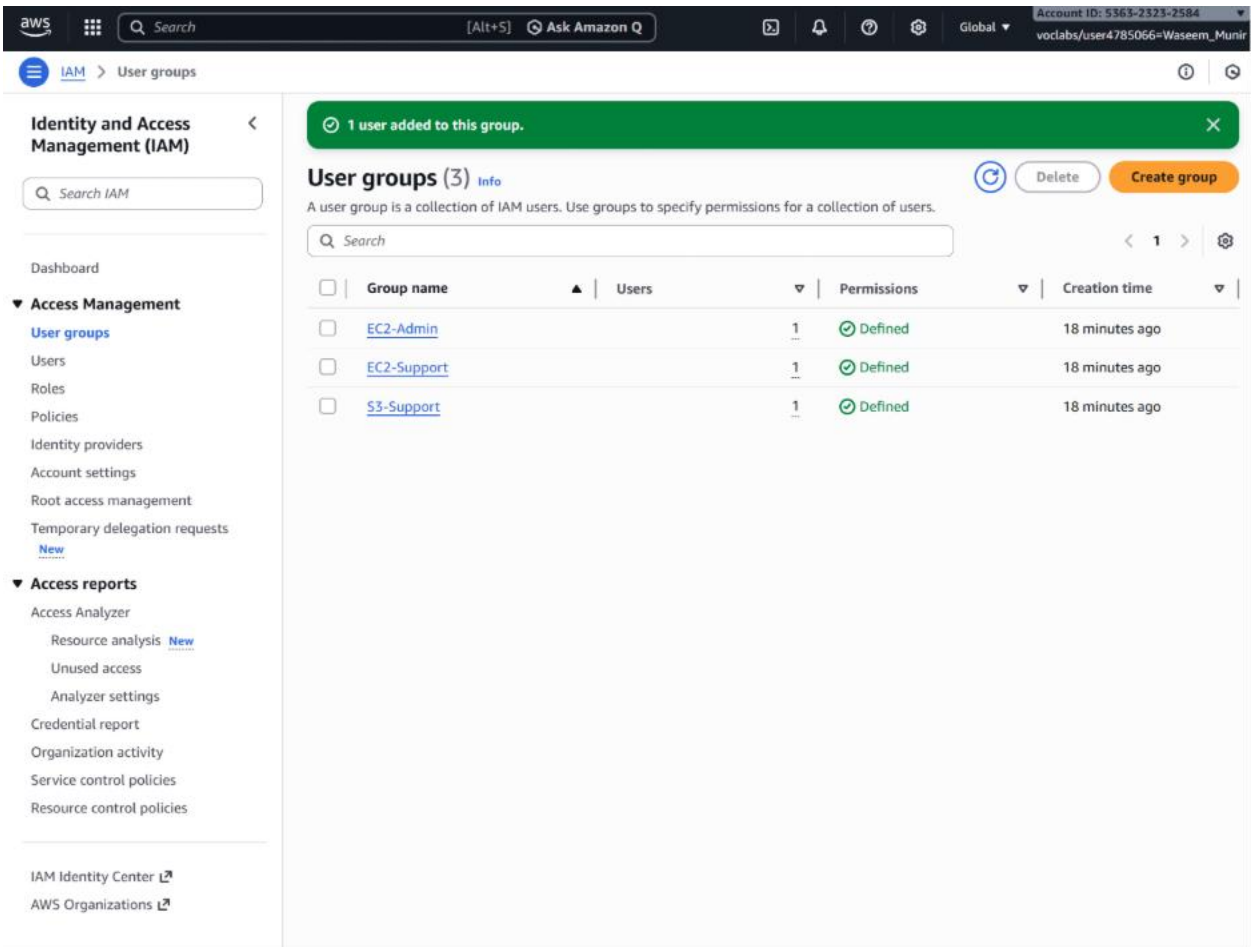


17. Using the similar steps above, add user-2 to the EC2-Support group

18. Using the similar steps above, add user-3 to the EC2-Admin group

Note: *Each Group should now have a 1 in the Users column, indicating the number of Users in each Group.*

If you do not have a 1 beside each group, revisit the above instructions above to ensure that each user is assigned to a User group, as shown in the table in the Business Scenario section



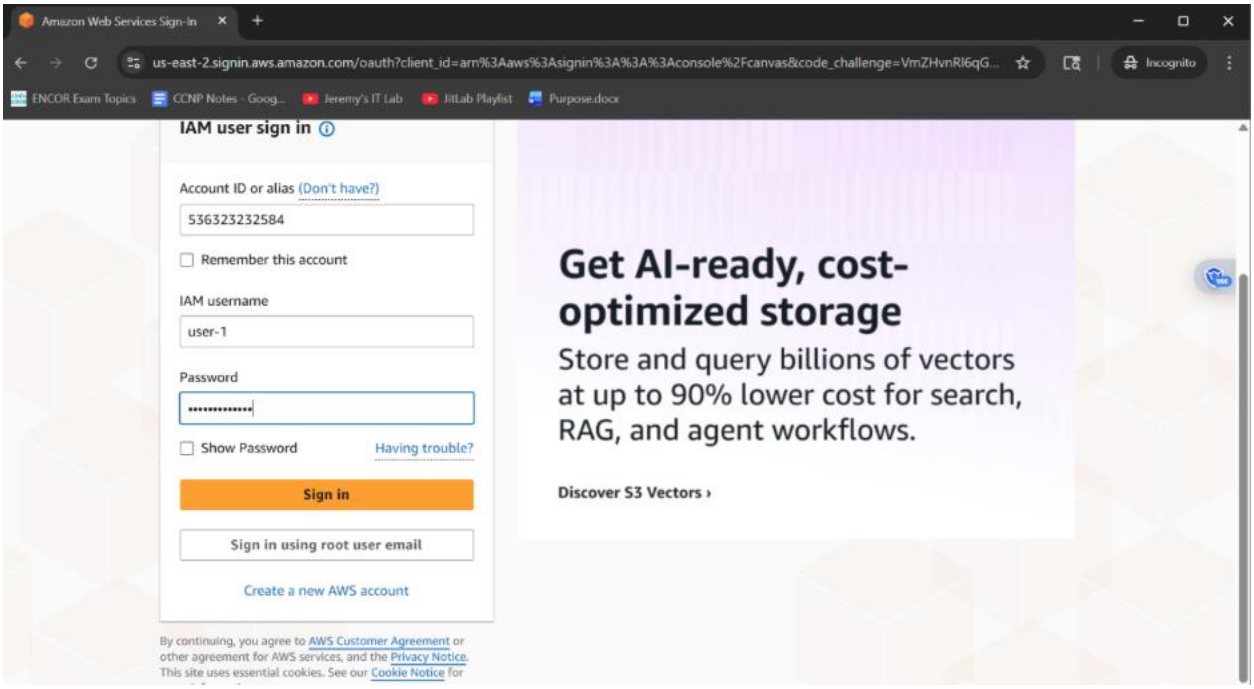
19. Choose “Dashboard” in the navigation pane on the left

20. Copy the Sign-in URL displayed on the right

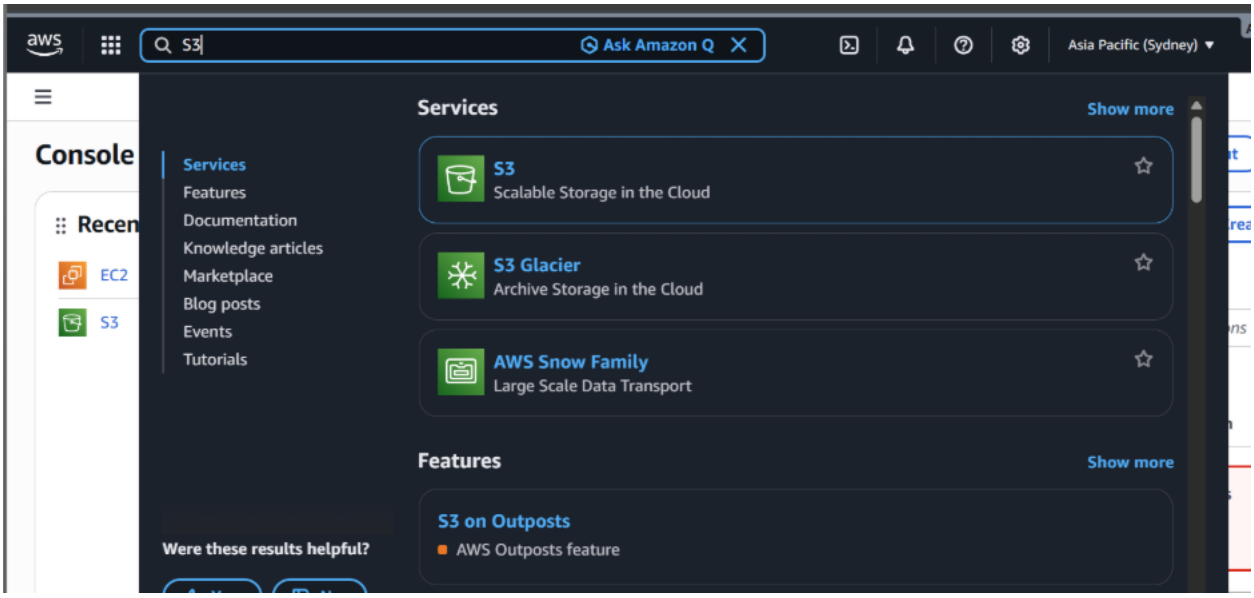
21. Open a private (Incognito) window & paste the link into the browser and go.

22. Sign in with

- IAM Username: user-1
- Password: Lab-Password1

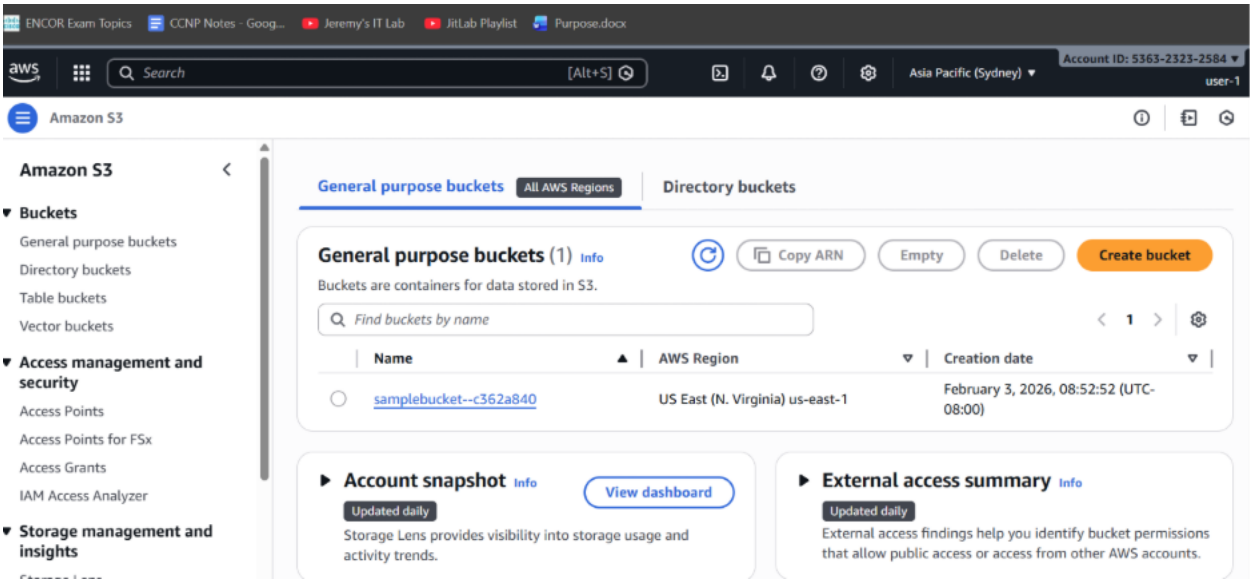


23. Search "S3" and click on it



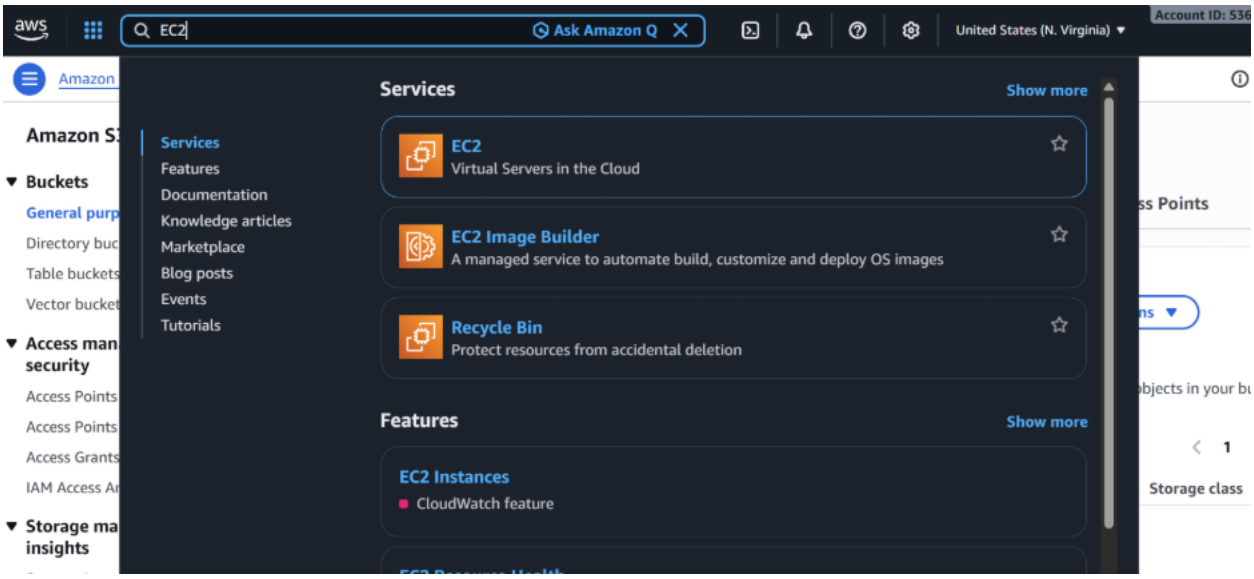
24. Click the name of the bucket and browse the contents

Note: *Since your user is part of the S3-Support Group in IAM, they have permission to view a list of Amazon S3 buckets and the contents.*



25. Search for and click on “EC2” in Services, then “Instances”

Note: *You cannot see any instances. Instead, you see a message that states You are not authorized to perform this operation. This is because this user has not been granted any permissions to access Amazon EC2.*



26. Sign out of user-1 by doing the following:
    - Choose “user-1” in the upper right corner
    - Choose “Sign Out”

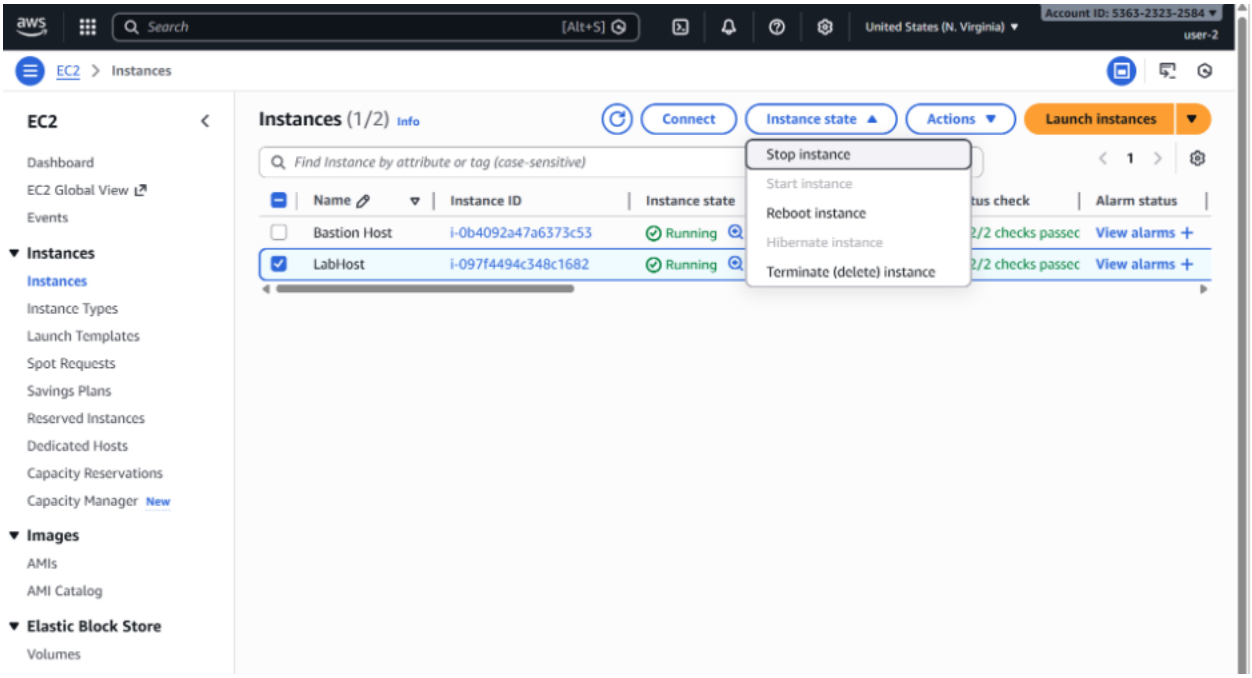
27. Paste the IAM sign-in link into the private browser tab’s address bar and enter

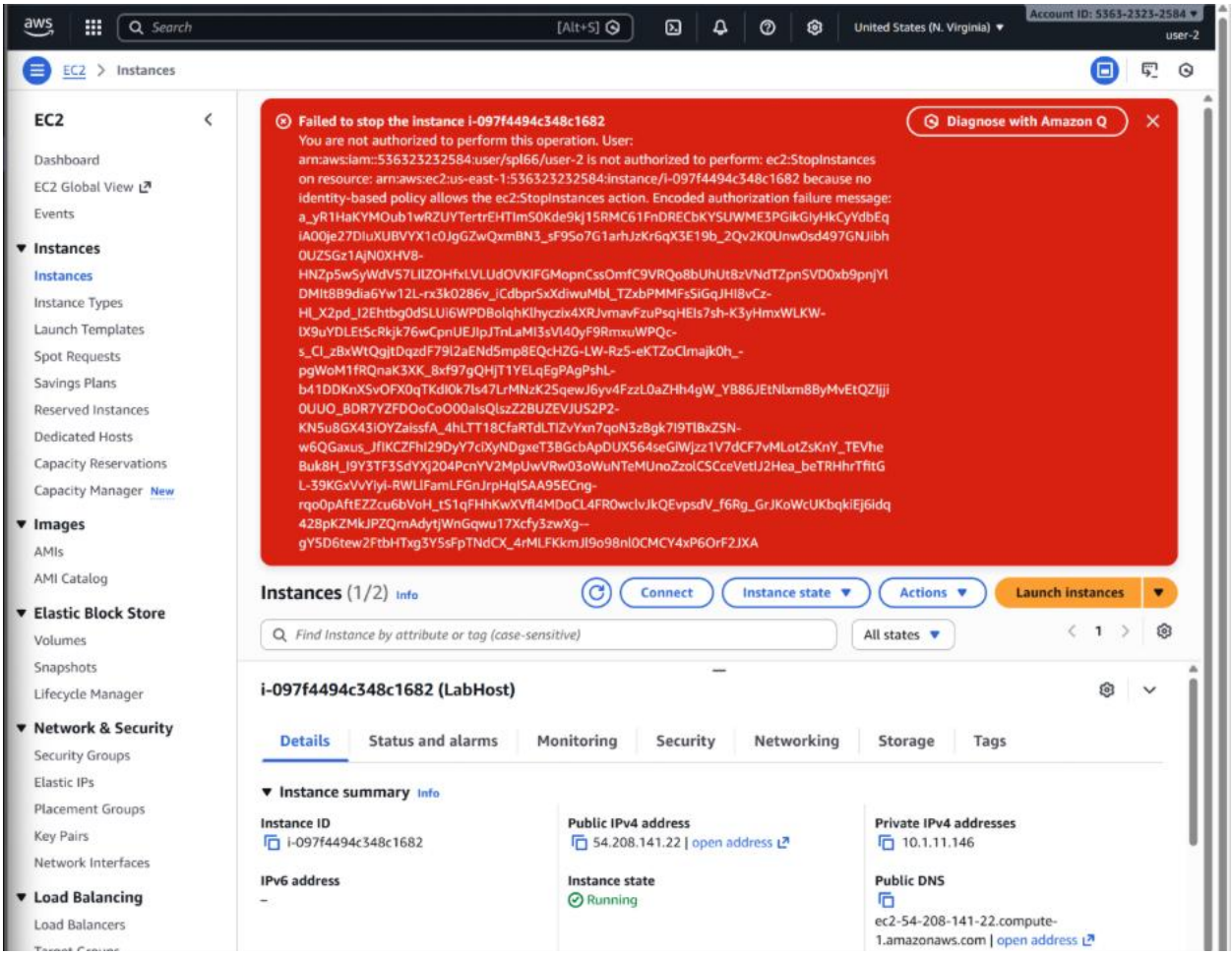
28. Sign in with:
    - IAM Username: user-2
    - Password: Lab-Password2

29. Search and choose “EC2” in the search bar

30. Choose “Instances”

Note: *You will notice that you can SEE the instance, but you cannot CHANGE the instance. This is because the IAM user has Read Only permissions. If you click on “Stop instance” for example, you will receive an error message prohibiting you from doing it.*





31. Paste the IAM sign-in link to the private window and hit Enter

32. Sign in with
    - IAM Username: user-3
    - Password: Lab-Password3

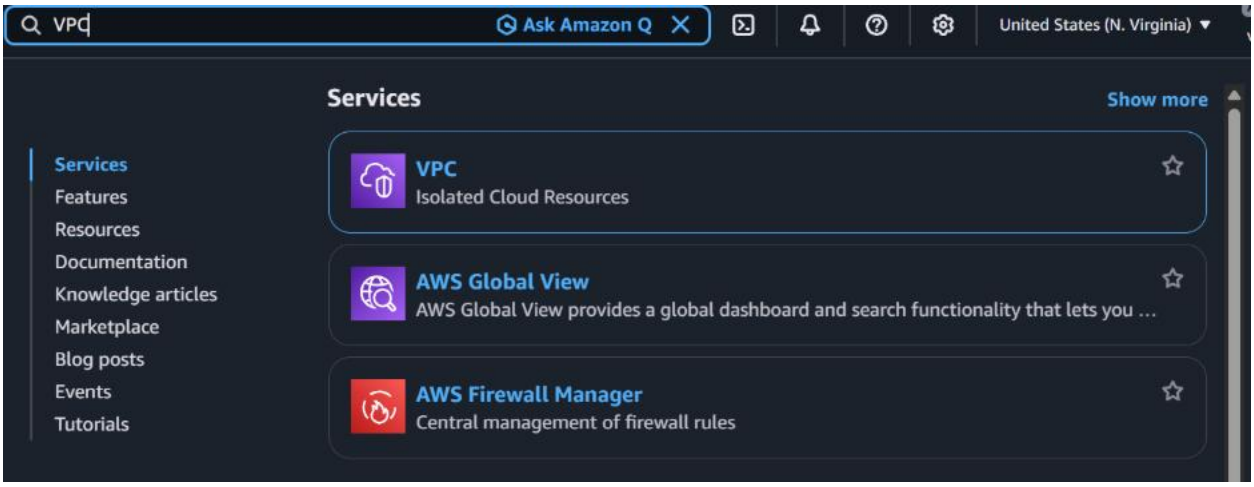
33. Search and choose “EC2” in the search bar, and click “Instances”

34. Select “LabHost” and click “Stop Instance” in the Instance state menu

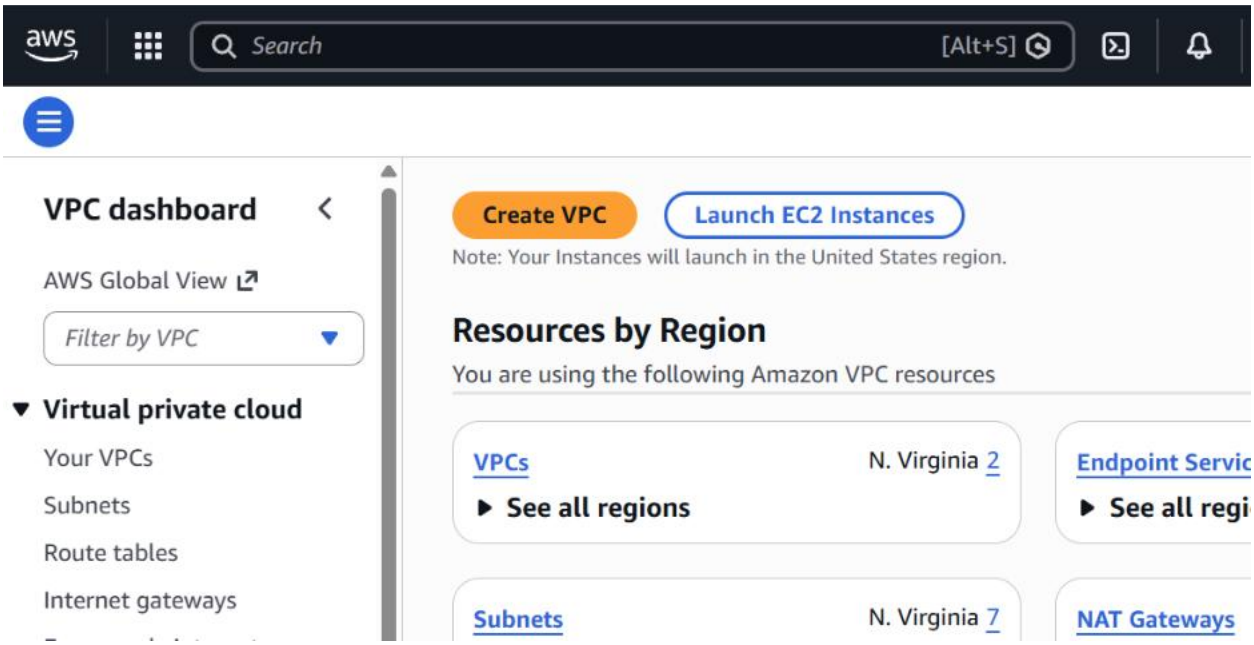
Note: The instance will stop because you have EC2 Administration permissions

AWS Lab 2: Virtual Private Cloud

1. In the search box to the right of Services, search for and choose VPC to open the VPC console.



2. Choose VPC Dashboard and click “Create VPC”



3. Configure with the following details

- Choose **VPC and more**.
- Under **Name tag auto-generation**, keep *Auto-generate* selected, however change the value from project to `lab`.
- Keep the **IPv4 CIDR block** set to 10.0.0.0/16
- For **Number of Availability Zones**, choose **1**.
- For **Number of *public* subnets**, keep the **1** setting.
- For **Number of *private* subnets**, keep the **1** setting.
- Expand the **Customize subnets CIDR blocks** section
  - Change **Public subnet CIDR block in us-east-1a** to `10.0.0.0/24`
  - Change **Private subnet CIDR block in us-east-1a** to `10.0.1.0/24`
- Set **NAT gateways** to **In 1 AZ**.
- Set **VPC endpoints** to **None**.
- Keep both **DNS hostnames** and **DNS resolution** *enabled*.

4. Confirm the settings you have configured

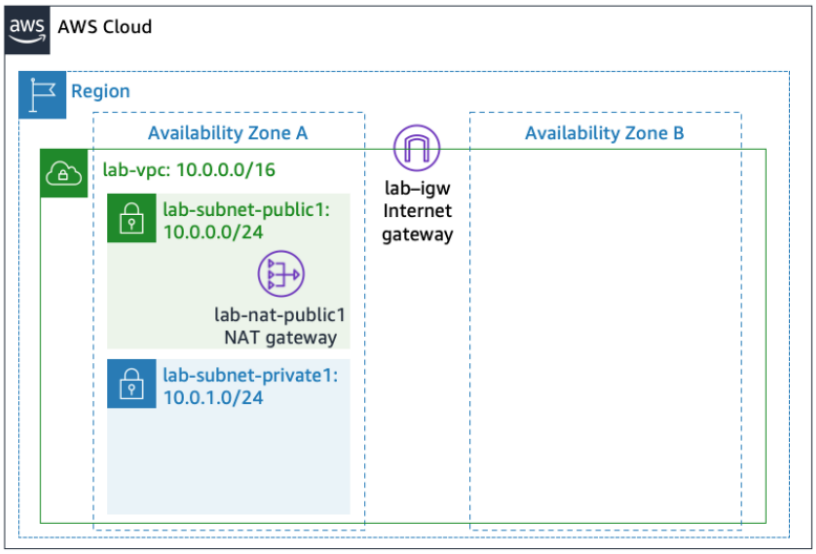
- **VPC:** `lab-vpc`
- **Subnets**:
  - us-east-1a
    - ***Public* subnet name:** `lab-subnet-public1-us-east-1a`
    - ***Private* subnet name:** `lab-subnet-private1-us-east-1a`
- **Route tables**
  - `lab-rtb-public`
  - `lab-rtb-private1-us-east-1a`
- **Network connections**
  - `lab-igw`
  - `lab-nat-public1-us-east-1a`

5. Choose “Create VPC” at the bottom of the screen. Then click “View VPC.”

Explanatory Note:

The wizard has provisioned a VPC with a public subnet and a private subnet in one Availability Zone with route tables for each subnet. It also created an Internet Gateway and a NAT Gateway.

To view the settings of these resources, browse through the VPC console links that display the resource details. For example, choose Subnets to view the subnet details and choose Route tables to view the route table details. The diagram below summarizes the VPC resources you have just created and how they are configured.



Public Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet gateway

Private Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	NAT gateway

An Internet gateway is a VPC resource that allows communication between EC2 instances in your VPC and the Internet.

The lab-subnet-public1-us-east-1a public subnet has a CIDR of 10.0.0.0/24, which means that it contains all IP addresses starting with 10.0.0.x. The fact the route table associated with this public subnet routes 0.0.0.0/0 network traffic to the internet gateway is what makes it a public subnet.

A NAT Gateway is a VPC resource used to provide internet connectivity to any EC2 instances running in private subnets in the VPC without those EC2 instances needing to have a direct connection to the internet gateway.

The lab-subnet-private1-us-east-1a private subnet has a CIDR of 10.0.1.0/24, which means that it contains all IP addresses starting with 10.0.1.x.

6. Choose “Subnets” in the left navigation pane

7. Choose “Create subnet” then configure with the following:

- **VPC ID: lab-vpc** (select from the menu).
- **Subnet name:** `lab-subnet-public2`
- **Availability Zone:** Select the *second* Availability Zone (for example, us-east-1b)
- **IPv4 CIDR block:** `10.0.2.0/24`

Note: *Confirm all IP addresses start with 10.0.2.x inside the subnet.*

8. Choose “Create Subnet”

Note: *The second public subnet was created. You will now create a second, private, subnet.*

9. Choose “Create subnet” then configure:

- **VPC ID:** `lab-vpc`
- **Subnet name:** `lab-subnet-private2`
- **Availability Zone:** Select the *second* Availability Zone (for example, us-east-1b)
- **IPv4 CIDR block:** `10.0.3.0/24`

Note: *Confirm all IP addresses start with 10.0.3.x inside the subnet.*

10. Choose “Create Subnet,” then “Route Tables”

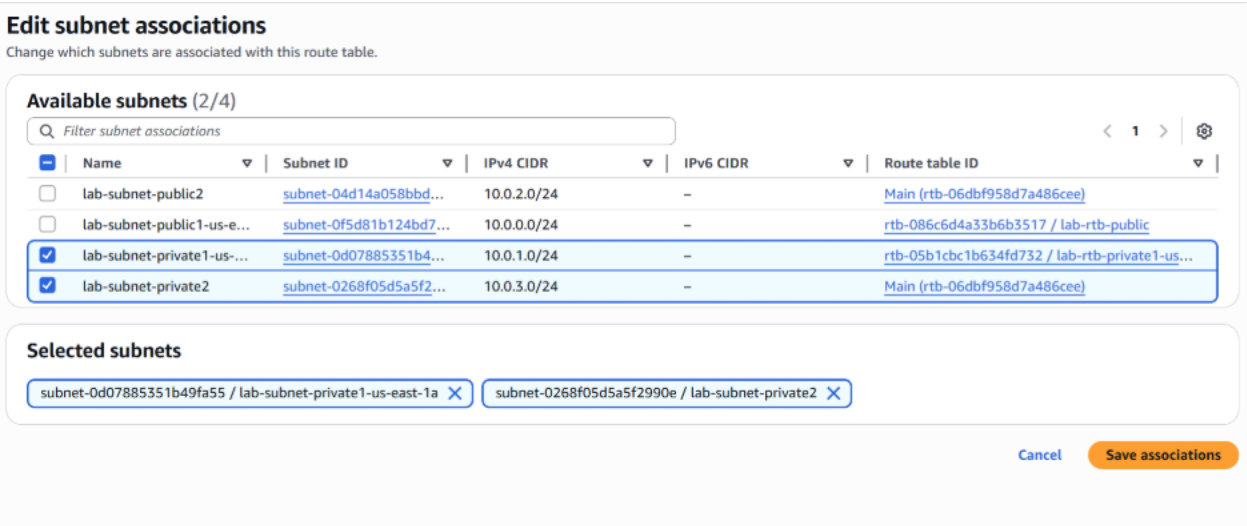
11. Select the checkmark next to “lab-rtb-private1-us-east-1a" route table

12. Click “Routes” in the lower pane

13. Choose “Subnet Associations”

14. Choose “Edit subnet associations”

15. Leave lab-subnet-private1-us-east-1a selected and click lab-subnet-private2



16. Click “Save associations”

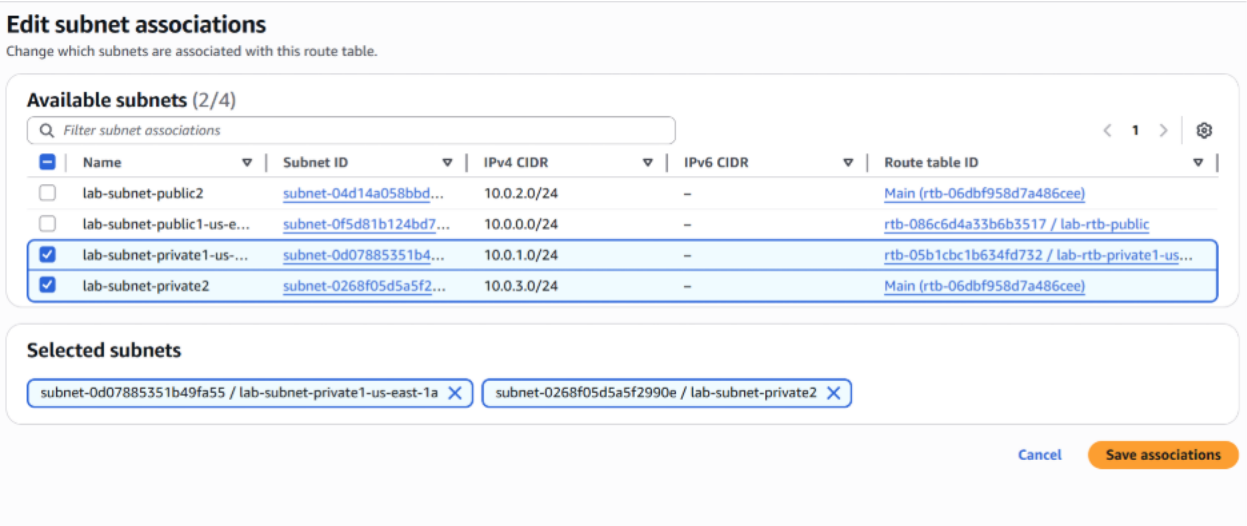
17. Select the “lab-rtb-public" route table

18. Click on “Routes”

19. Choose “Subnet Associations”

20. Choose “Edit subnet associations”

21. Leave lab-subnet-public1-us-east-1a selected and click lab-subnet-public2



22. Click “Save associations”

23. Click “Security Groups” on the left

24. Choose “Create security group” and then configure:

- **Security group name:** `Web Security Group`
- **Description:** `Enable HTTP access`
- **VPC:** choose the X to remove the currently selected VPC, then from the drop down list choose **lab-vpc**

25. In the inbound rules pane, choose “Add rule”

26. Configure the following settings:

- **Type:** *HTTP*
- **Source:** *Anywhere-IPv4*
- **Description:** `Permit web requests`

27. Scroll to the bottom and click “Create Security Group”

28. Search for and click “EC2”

29. Choose “Launch Instance,” and configure with the following settings:
    - Name: Web Server 1
    - Quick Start AMI: Amazon Linux
    - AMI: Amazon Linux 2023 AMI
    - Instance Type: t2.micro
    - Key Pair Name: vockey

30. Configure the Network Settings:
    - Network: lab-vpc

    - Subnet: lab-subnet-public2
    - Auto-assign public IP: Enable
- Then click “Select existing security group” under “Firewall (Security Groups)”
    - Select “Web Security Group” under “Common security groups”’

31. Expand the “Advanced Details” panel and copy paste the code below into “User Data”

```
#!/bin/bash

# Install Apache Web Server and PHP

dnf install -y httpd wget php mariadb105-server

# Download Lab files

wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip

unzip lab-app.zip -d /var/www/html/

# Turn on web server

chkconfig httpd on

service httpd start
```

Note: *This script will run with root user permissions on the guest OS of the instance. It will run automatically when the instance launches for the first time. The script installs a web server, a database, and PHP libraries, and then it downloads and installs a PHP web application on the web server.*

32. Click “Launch Instance”

Note: *You should see a “Success Message.”*

33. Click “View all Instances” & wait until you see 2/2 checks passed

34. Select “Web Server 1”

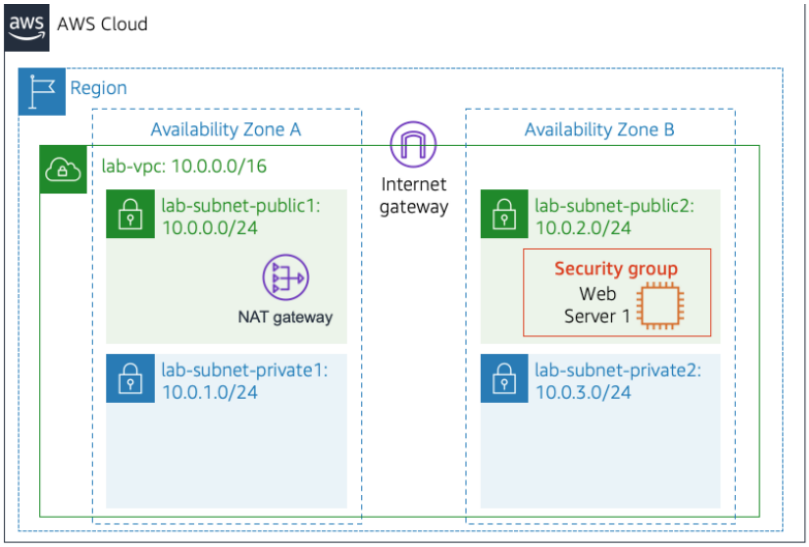
35. Copy the Public IPv4 DNS value in the Details tab

36. Open a new browser tab, and paste the Public DNS value

Meta-Data	Value
InstanceId	*i-033ed48e589b3ccc5*
Availability Zone	*us-east-1b*

Current CPU Load: **1%**

Note: *You should see a web page displaying the AWS logo and instance meta-data values. The complete architecture you deployed is:*



Public Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet gateway

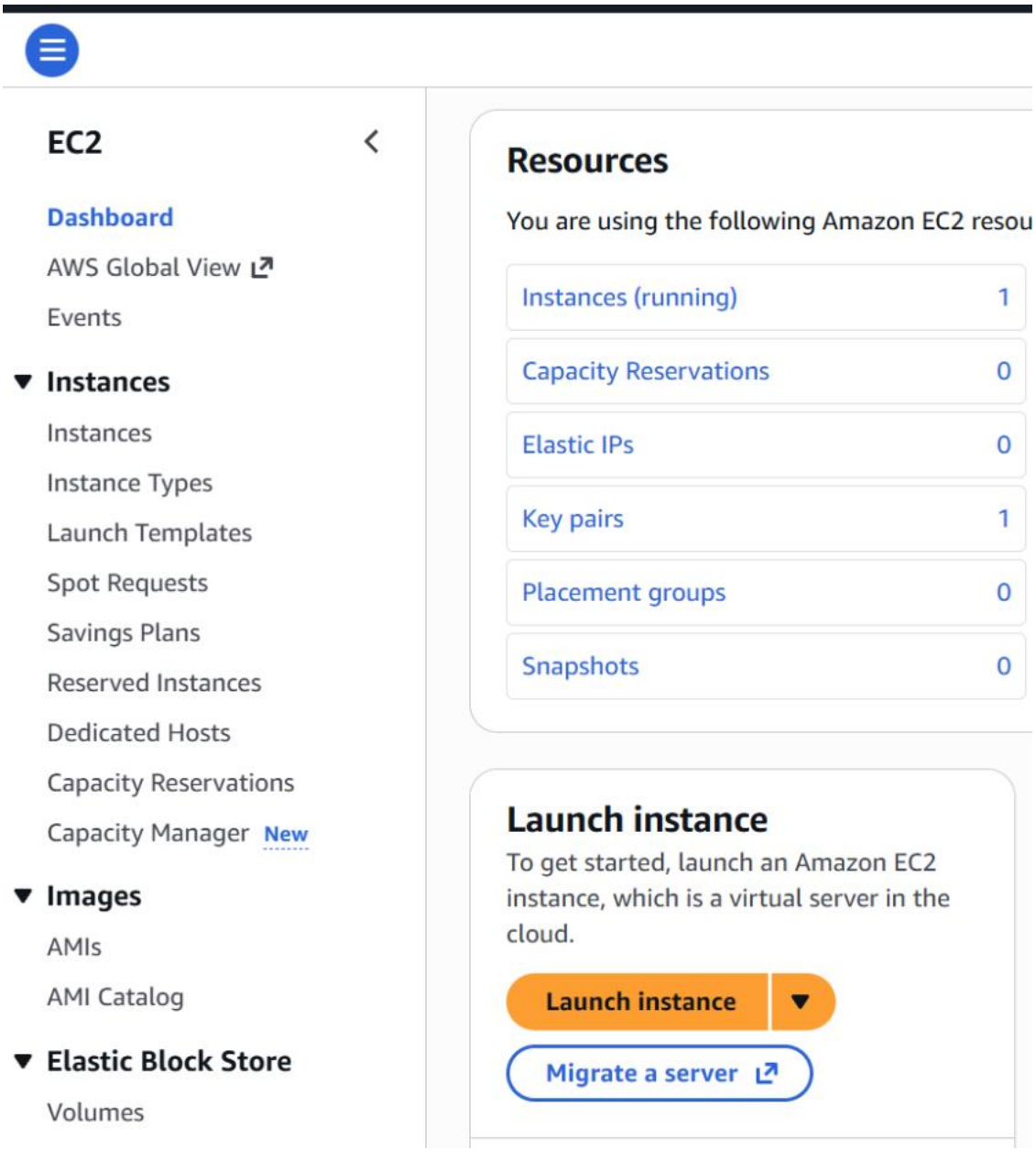
Private Route Tables

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	NAT gateway

Lab 3: Intro to Amazon EC2

1. Choose “Services” then “Compute” then “EC2”

2. Choose the “Launch Instance” menu and select “Launch Instance”

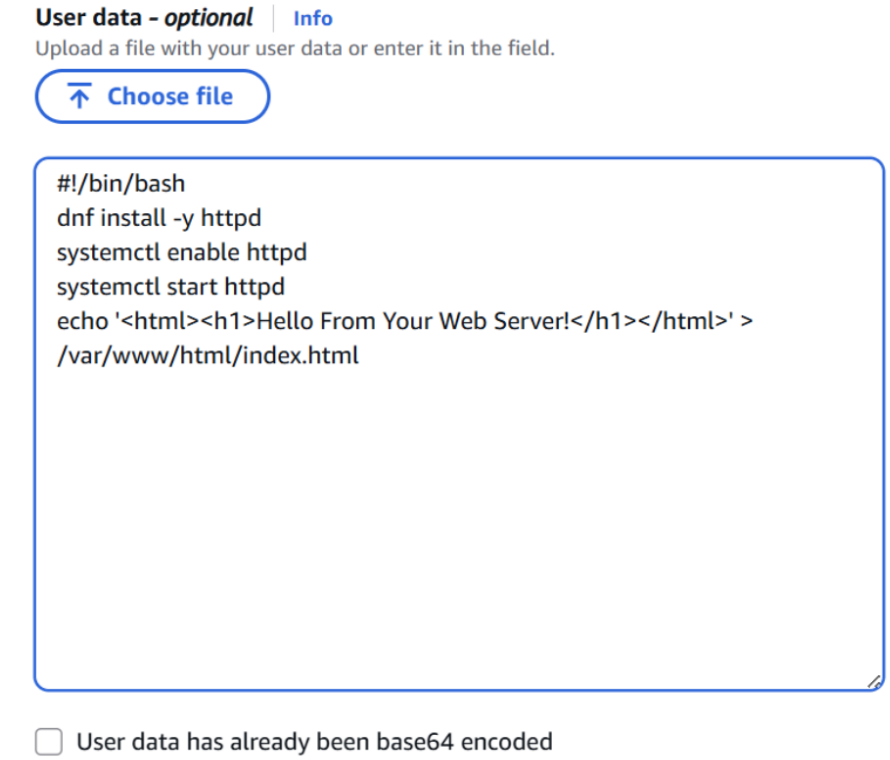


3. Configure the instance with the following settings
    - Name: Web Server
    - Quick Start AMIs: Amazon Linux
    - AMI: Amazon Linux 2023 AMI
    - Instance type: t2.micro
    - Key pair name: vockey
        - Click “Edit” next to “Network Settings”
    - VPC: Lab VPC
        - Click “Create security group” under “Firewall (security groups)”
    - Security Group name: Web Server Security Group

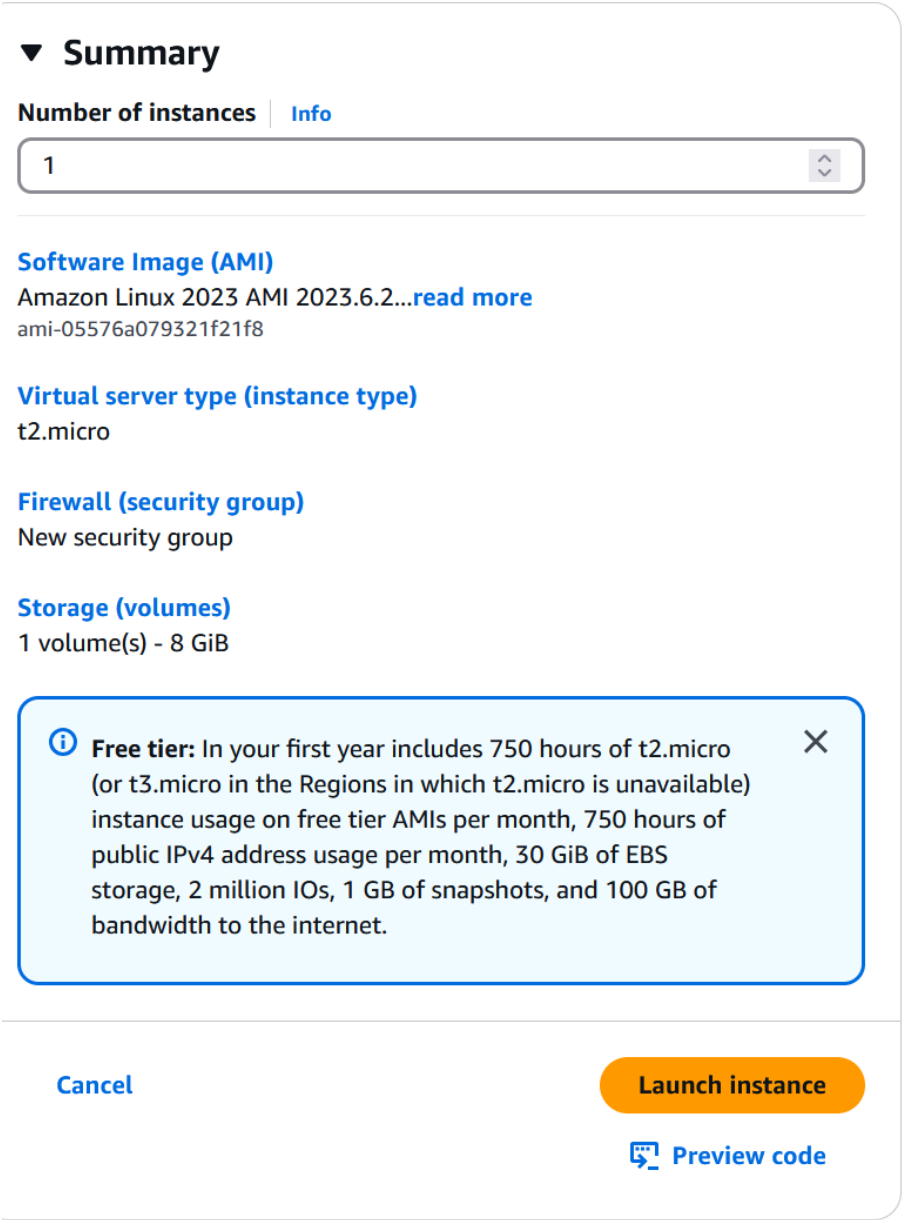
- Remove the rule under “Inbound security group rules”
- Description: Security group for my web server
    - Click & Expand “*Advanced Details*”
- Enable “Termination Protection”

- Scroll to the bottom of the page and paste the following in “User Data”:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' >
/var/www/html/index.html
```



4. Choose “Launch Instance” under “Summary”



Note: *You should see a “Success” message*

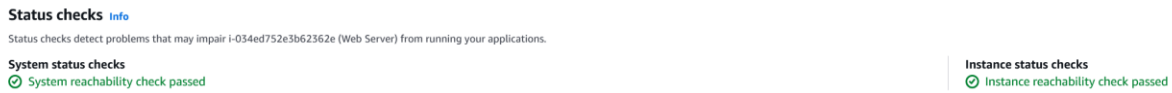
5. Choose “View All Instances”

6. Select “Web Server”

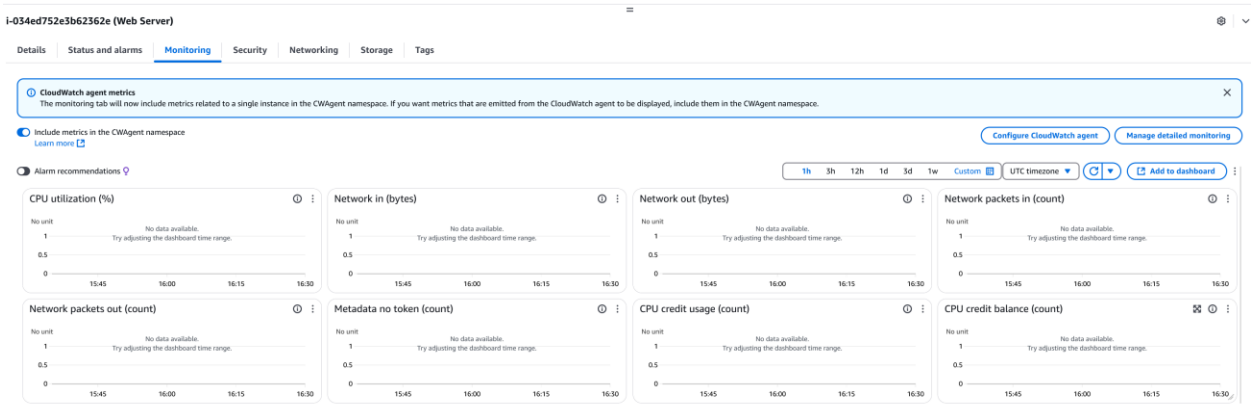
7. Wait for the instance to display the following:
   - Instance State: Running
   - Status Checks: 2/2 checks passed

Note: *Congratulations, you have launched an EC2 Instance!*

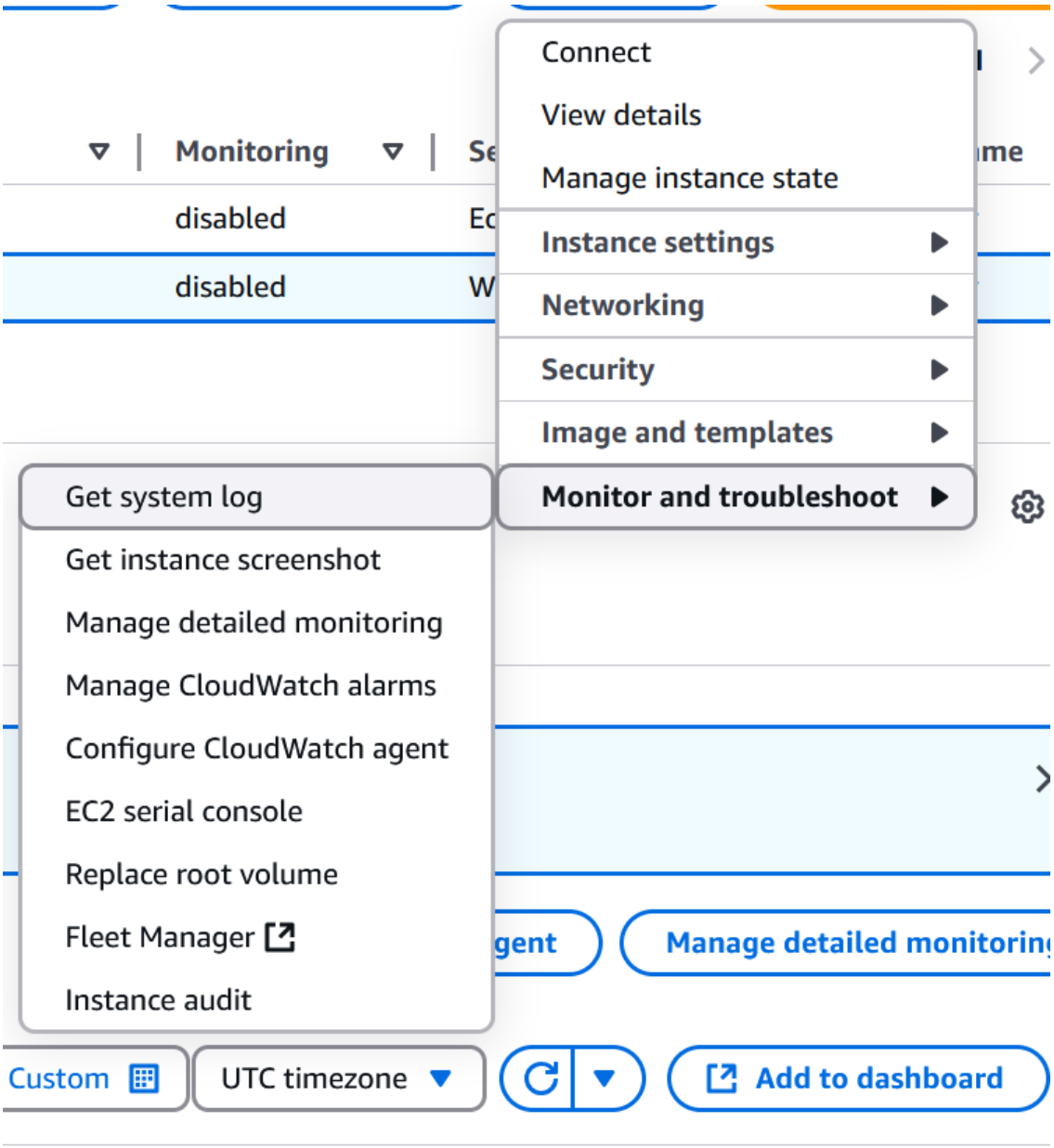
8. Click “Status Checks” tab



9. Choose “Monitoring”



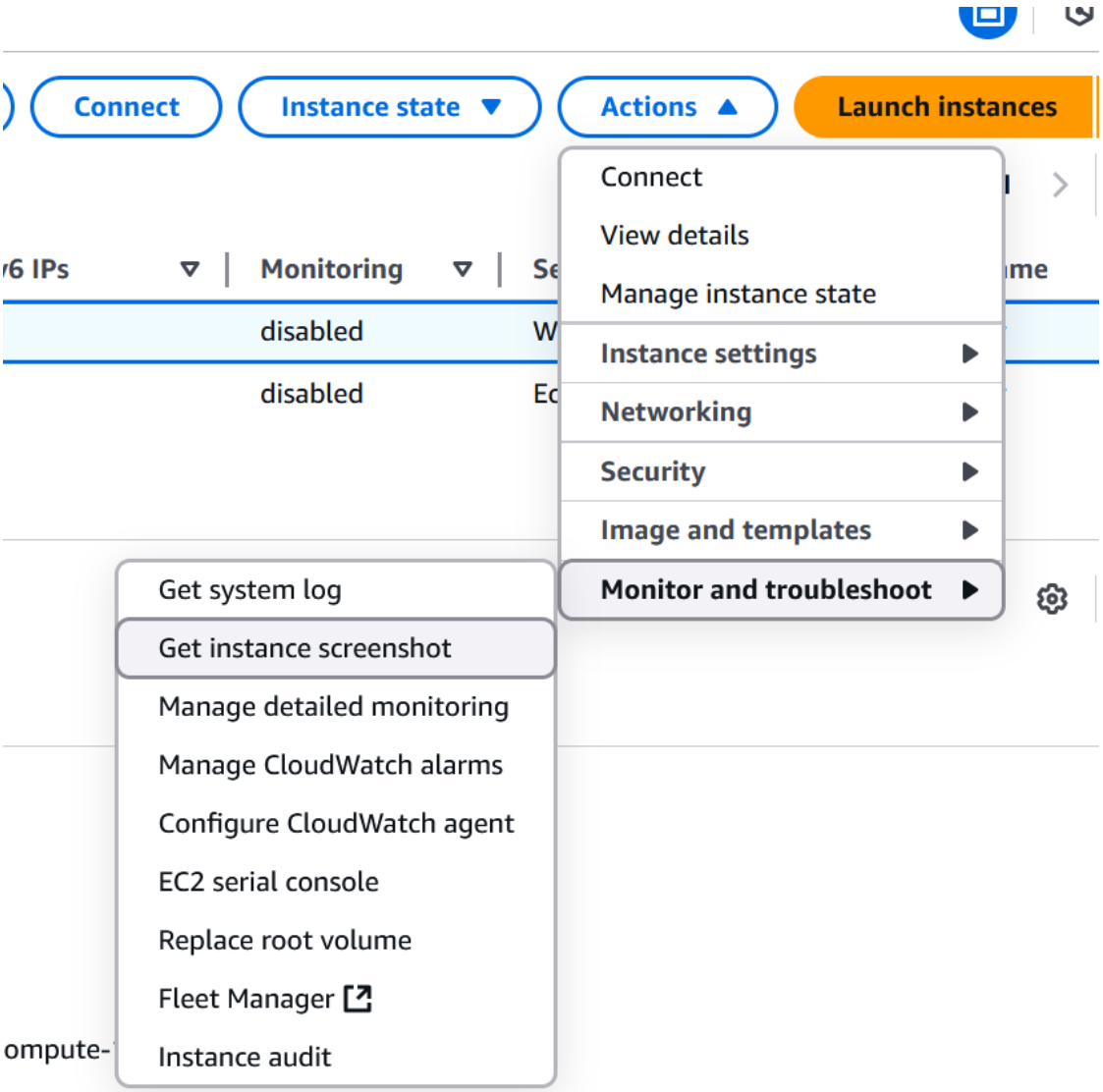
10. Select “Monitor & Troubleshoot -> Get system log” under “Actions” menu



11. Scroll through your output and note that your HTTP package was installed from the user data

12. Choose “Cancel”

13. Ensure that “Web Server” is still selected. Then click “Get Instance Screenshot” under “Monitor & Troubleshoot” in the “Actions” menu

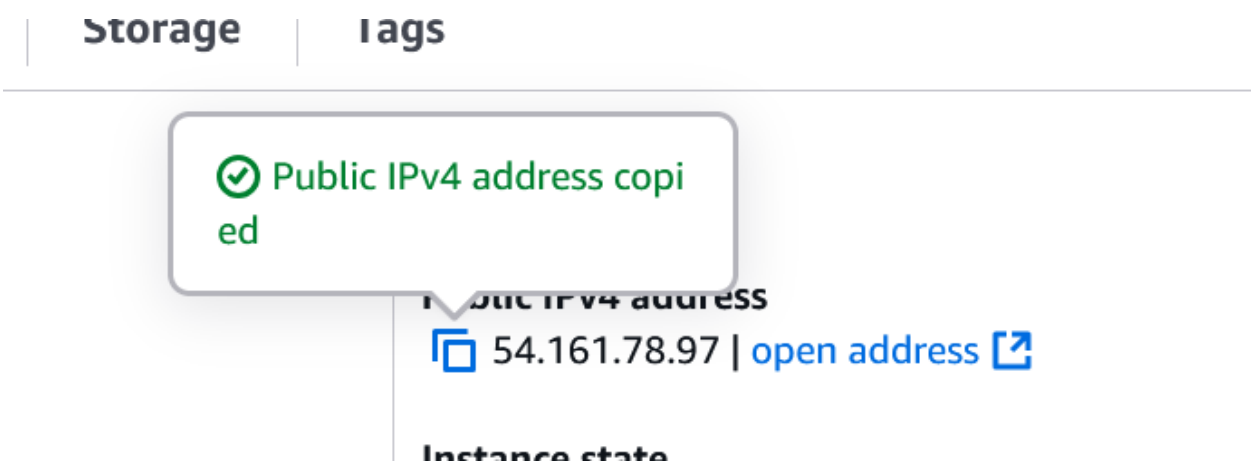


14. Click “Cancel”

Note: *Congratulations, you have explored several ways to monitor your instance*

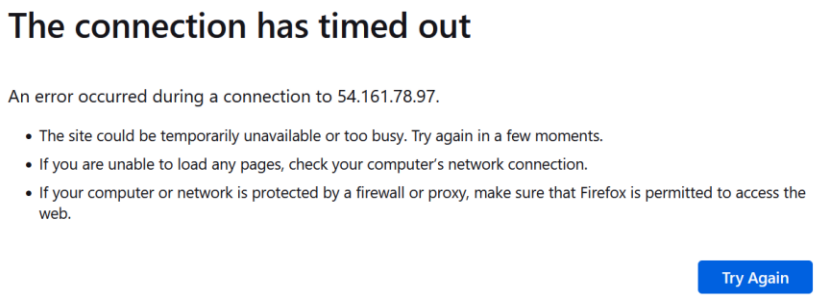
15. Ensure that “Web Server” is selected. Click “Details”

16. Copy the Public IPv4 address of your instance to your clipboard



17. Enter this IP Address in a new web browser and press “Enter”

Note: *You cannot access your web server because your security group does not permit inbound traffic on port 80; the port used for HTTP web requests.*

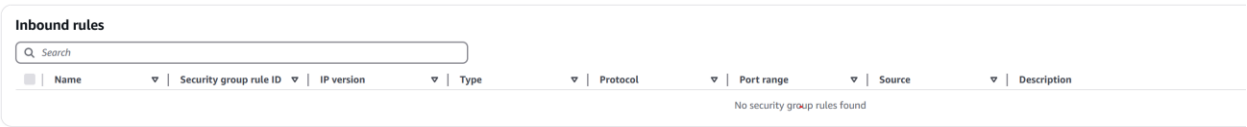


18. Keep the browser tab open, but return to the “EC2 Console” tab

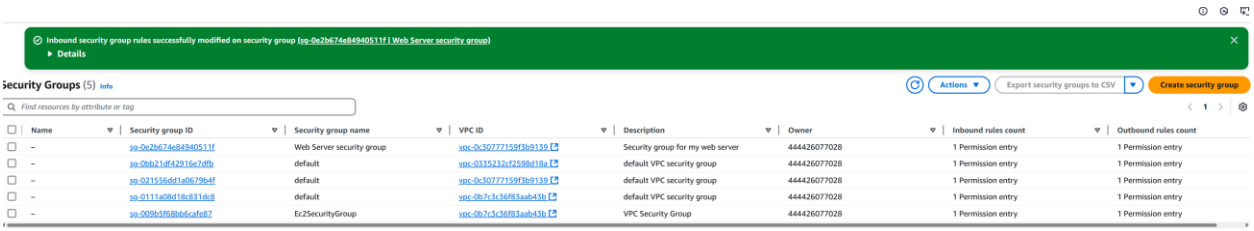
19. Choose “Security Groups”

20. Select “Web Server Security Group”

21. Choose “Inbound Rules” tab

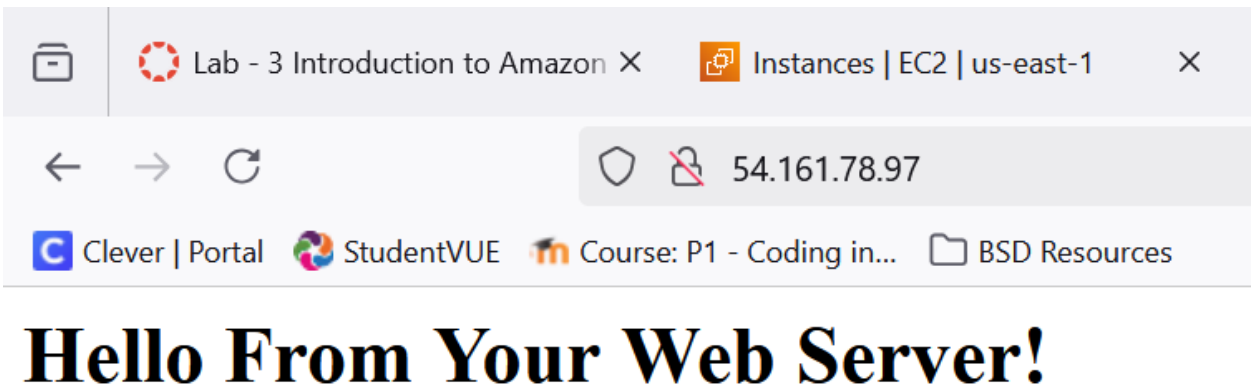


22. Choose “Edit inbound rules,” select “Add rule,” and configure
    - Type: HTTP
    - Source: Anywhere-IPv4
    - Choose “Save Rules”



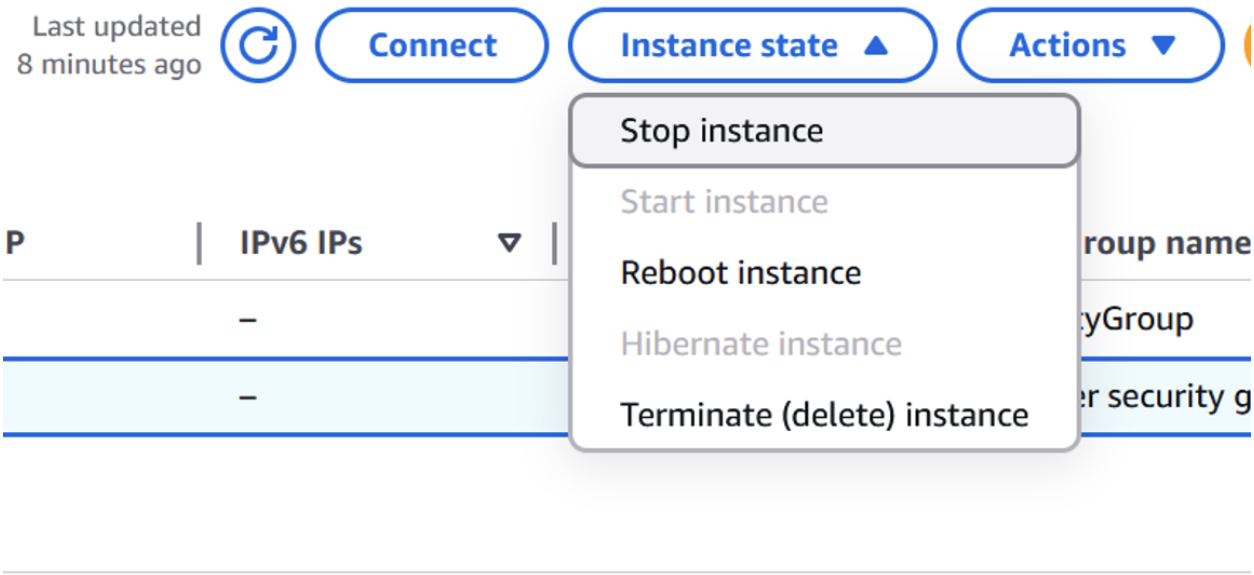
23. Refresh the web server tab you previously opened and refresh the page.

Note: *You should see the message “Hello From Your Web Server”*



24. Click “Instances” and select “Web Server” in the “EC2 Management Console”

25. Select “Stop Instance” in the “Instance State” menu



26. Choose “Stop” and wait for it to display: Stopped

27. Select the Web Server instance, then in the “Actions” menu, click “Change Instance Type” under the “Instance Settings” menu. Configure the following:
    - Instance Type: t2.small
    - Choose “Apply”

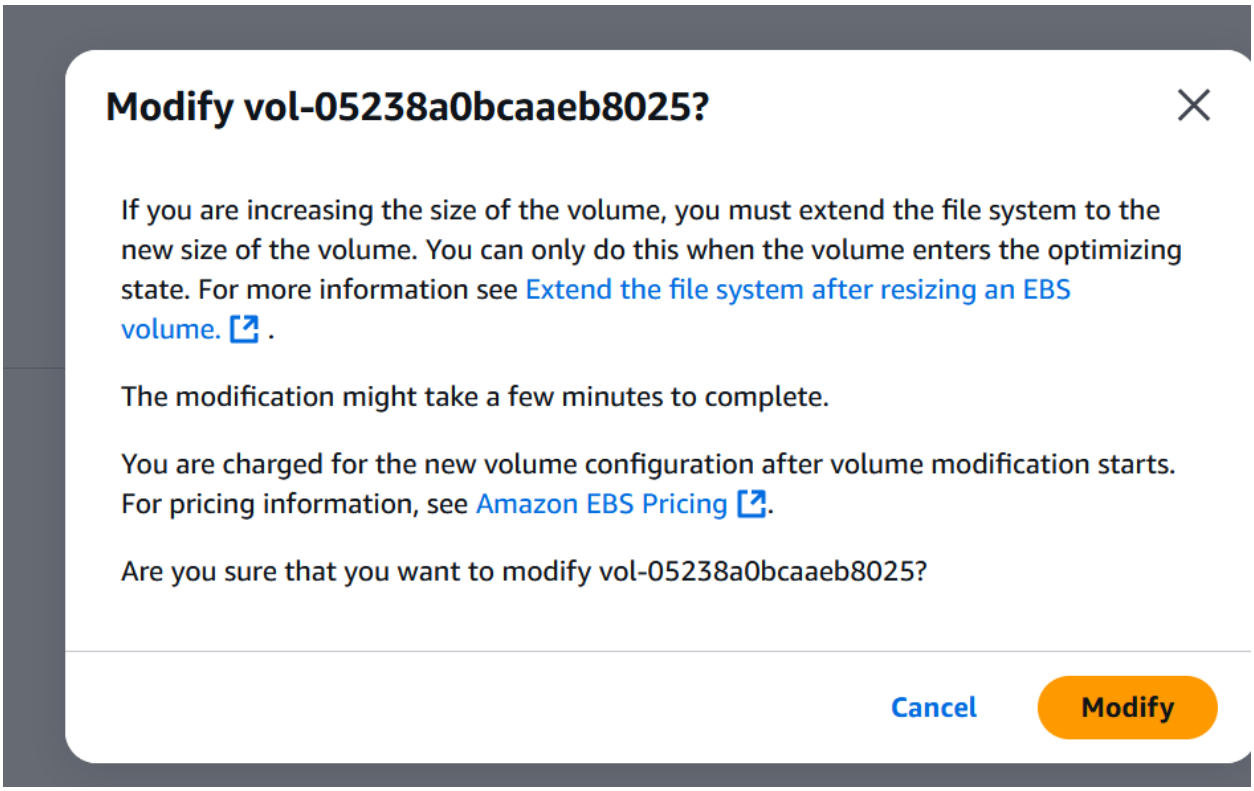
28. Select “Change Stop Protection” under “Instance Setting.” Select “Enable,” then “Save”

29. Choose the “Storage” tab, select the name of the Volume ID, and select the checkbox next to the volume that it displays.

30. In the “Actions” menu, select “Modify volume.”

31. Change the size to “10”

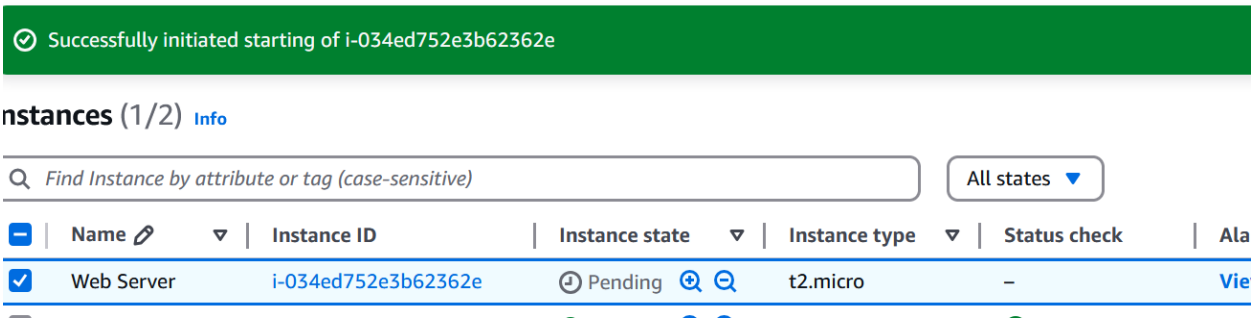
32. Choose “Modify” then confirm by clicking “Modify” again.



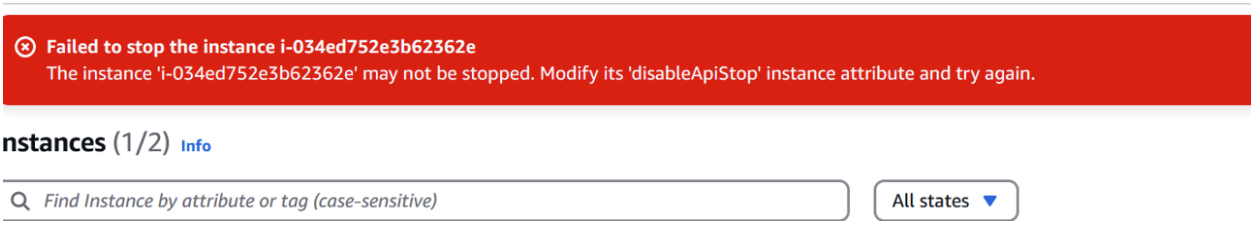
33. Click “Instances” in the left navigation pane & select the Web Server Instance

34. Select “Start Instance” in the “Instance State” menu

Note: *Congratulations, you have resized your Amazon EC2 Instance.*



35. Now in the “Instance State” menu, click “Stop Instance,” then click “Stop.”



Note: *The stop protection we enabled earlier will give you an error to stop you from deleting the instance.*

Problems

1. **Lack Of Knowledge**

This lab had mostly clear instructions and had very little issues with it. The one issue I had was a lack of knowledge about the general subjects being discussed in the lab at first, which made it more difficult to conceptualize. I was able to overcome this by studying for my AWS Cloud Practitioner Exam using Boson to understand the concepts better.

Conclusion:

This lab was great at teaching the basic introductory concepts of AWS, and it was a great way to learn how these concepts come together in the actual AWS interface.